

City Air Makes You Free

(in some countries, or if you are upper class or financially satisfied)

Friday 2nd October, 2026 10:07

target jou:

cities if8.2,

ariq if3ish,

some soc j have soc lit like Toennies simmel maybe even durkheim maybe sth like j or urb soc

We use World Values Survey 7 (2017-2023) across 64 countries to test the hypothesis that cities are more free. Mostly we find the opposite: in ?? out ?? countries there is mostly no relationship or even in ?? cases positive relationship. Still, in ?? countries cities are indeed more free. Within-country urban-rural freedom/autonomy differences are mostly insignificant, but some effect sizes tend to be substantial, about 0.3-0.6 on 1-10 freedom/autonomy scale. Another key finding is that, as already theoretized by Toennies, the upper classes (and financially satisfied) are “active and alive” in cities or as we find here free/autonomous. And lower class, financially dissatisfied, and the poor are more free/autonomous outside of the cities.

WORLD VALUES SURVEY, FREEDOM, AUTONOMY, URBANISM, URBAN, RURAL, URBAN-RURAL, CITIES

City life and Gesellschaft down the common people to decay and death. Ferdinand Toennies

In the city and therefore, where general conditions characteristic of the Gesellschaft prevail, only the upper strata, the rich and the cultured, are really active and alive. Ferdinand Toennies

City air makes you free¹—the saying originated in the Middle Ages, and it meant freedom from feudalism in cities, non-feudal islands in a sea of feudalism (Harvey 2012). Feudalism has long been gone, but we may actually be living under feudalism again, so called techno-feudalism (Varoufakis 2024, Morozov 2022). While modern cities aren't clearly more free from techno-feudalism,² as Middle Ages cities used to be more free from feudalism, cities are still often seen in the literature and popular culture as more free than ruralities. Hence, it'd be already a finding worth reporting that people don't have more freedom/autonomy in cities, because often the expectation is that they do.

Freedom/autonomy is an end-in-itself, a desirable outcome in just about any respect, except perhaps responsibility. In the Middle Ages there was little freedom, but also little responsibility—for instance the poor and the slaves were not responsible for their lot—their situation was God given. Now the poor have to worry about being poor and about it being their fault—everyone is free after all and their achievements and experiences are of their own making.³

1 Freedom/autonomy defined

Some authors use term “freedom” (Okulicz-Kozaryn 2014) and some “autonomy” (Steckermeier 2022)—we use the terms interchangeably, but prefer “freedom” as it seems to sound and fit better in this context.

¹(Stadt Luft macht frei) (Park et al. [1925] 1984, p. 12)

²Cloud fiefdoms, rent, and digital serfdom through platforms like Google and Amazon “enslaves” urbanites and ruralities, but still perhaps city may be more free in a sense that there is more choice and opportunity say with local traditional on the ground commerce.

³For instance, people in the US feel relatively free and so they blame the poor for their lot, while Poles feel less free and also blame the poor less.

What is freedom/autonomy?⁴ To be free is to be able to choose, to control and direct one's own life. Freedom is having control—a person is free if she is her own master. It could be labeled “psychological freedom:” “Freedom means a degree of harmony between basic motives and overt behavior” (Bay 1970, p.83 cited in Okulicz-Kozaryn 2014). It is “capability to choose,” “awareness of alternatives” and “courage to choose” (Veenhoven 2000).⁵ Clearly, there are more choices in city v ruralites, so at least in a sense of choice there is more freedom. But then too much choice may also backfire and even lead to a paralysis (Schwartz 2004).

Freedom/autonomy is a broad concept and could be sub-divided into dimensions—Hitlin and Long (2009) reviews the concept in sociology. See also a recent scale (D'Italia and Okulicz-Kozaryn 2025). Another useful review is Kouba (2016) and also relevant here as it points to our WVS measure.

2 Literature

There are theoretical reasons to expect more freedom in cities v ruralities. In rurality there is mechanical solidarity—rural folks function like parts of a simple machine—groups are held together through homogeneity and similarity: similar lifestyles, roles, shared values and morality (Durkheim [1893] 1997). Hence, ruralities force social conformity not conducive for individual freedom. On the other hand, organic solidarity in cities allows for more differentiation and individualism. Urbanites are also more tolerant (Tuch 1987). Likewise, there would be more *Gemeinschaft* in ruralities—more collective and social that may restrict personal freedom; and more *Gesellschaft* in cities that may promote individualism (but also hostility and aggression). Simmel observed that as urbanism overloads a person it forces to distance oneself from the environment (Simmel 1903), which also may be conducive for differentiation, individualism, and then perhaps freedom. On the other hand, urban hunkering down, withdrawal, and blasé attitude (Simmel 1903) are not agentic.

Many studies approach freedom as a predictor of subjective wellbeing, but freedom is also an end-in-itself (Okulicz-Kozaryn 2014), as recognized by Fromm (2013, 2012, 1992, 1964, 1944, [1941] 1994) and Sen (2000). The Universal Declaration of Human Rights states that freedom is a human right: freedom for the realization of one's human potential (art. 26, 29).

There are studies on freedom using WVS variable that we use in our study (e.g., Okulicz-Kozaryn 2014, Pitlik 2016, Nikolaev and Bennett 2016), but they do not examine urbanicity. There are many studies on the effect of urban-rural on subjective wellbeing (SWB) (e.g., Sørensen 2024, 2014, Knies 2026, Okulicz-Kozaryn and Valente 2021), but the effect of urban-rural on freedom has not really been studied.

TODO add here that paper leonie has in denmark that looks at urb-rur swb but also has in appendix freedom

While Ester and Vinken (1993) is not about urban-rural freedom specifically, it is indirectly related. Ester and Vinken (1993) uses similar dataset to ours, EVS/WVS, and tests yuppie value syndrome (yuppies are young urban professionals), and finds that yuppies tend to hold liberal social and ethical views, but are not: economically conservative, absent from politics, and don't have distinctive lifestyle emphasis.

Yamagishi et al. (2012) claims to test our “city free air” hypothesis, but Yamagishi et al. (2012) test is very different from ours—they only show that urbanites are more likely to chose extravagant pens than rural folks. Still, the study is useful in at least 2 ways. First they show that indeed there is more freedom in city (using extravagant pens test), and they provide theory and literature summarized in the next paragraph.

Cities provide freedom from traditional rural social constraints—*Gesellschaft* v *Gemeinschaft*—deviation from community norms is more likely to be reprimanded and ostracized in rural v urban. An urbanite may be more internally driven rather than externally constrained rural folk. City provides more opportunities for alternative jobs and social relations, so urbanite is more free by being less

⁴Here we focus on personal as opposed to societal freedom and follow Okulicz-Kozaryn (2014) in this paragraph.

⁵The degree of perceived control over choice is called “locus of control” in psychology and Verme (2009) argues that WVS item used here measures locus of control in addition to freedom.

concerned about social exclusion as she has plenty of alternatives. Also see discussion in Luca et al. (2023).

One mechanism is higher social pressure for conformity in ruralities—cities provide anonymity—indeed concern for moral reputation is greater in rural than urban areas (Danielson et al 2023 cited in Okulicz-Kozaryn 2026). The higher the population density, the lower the frequency of social comparison measuring social comparison with searches on Google (Okulicz-Kozaryn 2026).

But there is also theory suggesting less freedom in cities. City is stressed, hectic, anxious, materialistic, consumerist (Okulicz-Kozaryn 2015), and overstimulating (Simmel 1903, Lederbogen et al. 2011)—in such conditions freedom/autonomy is difficult. Steve Pile in his colorful writing argues that old cities can be a weight on its inhabitants with their history and past (Pile 2005a,b, Pile et al. 1999), and old cities carry melancholia (Pile 2005b). Weight of history and melancholia can inhibit how free one feels.

Perception of freedom can be counterintuitive and deceptive—a place may seem free, or indeed actively advertise itself as an oasis of freedom, but actually not be that free after all. The US advertises itself as “the land of the free” and yet people in the US feel less free at 7.7 than Mexicans and Colombians at above 8.1 (WVS Wave 7). Likewise, city may seem more free than ruralites, but as this paper finds, usually that's not the case. In a city v rural (or in the US v Latin America) there is often more social pressure (Okulicz-Kozaryn and Martinez 2025, Okulicz-Kozaryn et al. 2026)⁶ and so an urbanite is more of an “automaton” (to use Fromm's term) (Okulicz-Kozaryn 2014).

Also see appendix section AI literature review.

3 Hypotheses

H_1 : People in cities are freer than people in rural areas.

???? i would either use the full scale (pseudo-metric) or make a median split by country here

yeah its a mix some countries this or that, but most sig c yes more free in cities

H_2 : This difference is more pronounced in less developed countries

why we hypothesize this?

not much there neither iwth gdp nore inc group

but yes with perc urban—more urban more c, more freedom in city: in fig 13

H_3 : The “more” of freedom in cities mostly stems from rich people/upper classes (following the Toennies quote)

4 Data and model

World Values Survey (WVS) is a fantastic and free dataset available from <https://www.worldvaluessurvey.org>. It is a great source of data as compared to \$30k+ closed source Gallup data. It is helpful to use open source public data amidst corporatization of research and academia (Mills 2012a, Cox 2013, Mills 2012b, Catropa and Andrews 2020, Schmidlin 2015). WVS is representative of each country. Countries in the sample and sample sizes are shown in appendix.

Autonomy/freedom variable reads: “Some people feel they have completely free choice and control over their lives, while other people feel that what they do has no real effect on what happens to them. Please use this scale where 1 means 'none at all' and 10 means 'a great deal' to indicate how much freedom of choice and control you feel you have over the way your life turns out.”

⁶Also see a fantastic book by Frank (2012) on keeping up with the Joneses in the US.

Validity and reliability of the measure seem fine and the variable has been used successfully in the literature (e.g., Verme 2009, Inglehart et al. 2008, Veenhoven 2000, Okulicz-Kozaryn 2014) but still the question may be interpreted slightly differently in different languages/cultures, and some caution would be useful. This is also one reason for within-country analyses that we start with in the next section.

Urban-rural is a gradient, not a dichotomy (Okulicz-Kozaryn 2017), but here for simplicity we use 3 or 4 collapsed categories and focus on the endpoints of the spectrum: $> 500k$ v small $< 10k$ or $< 20k$ (separately for each country). $> 500k$ represents city—indeed it could even be bigger than 500k—urban unhappiness starts around several hundred thousand (Okulicz-Kozaryn 2017).

We dropped countries where there were fewer than 50 observations in cities $> 100k$. And in appendix as a robustness/sensitivity check with fewer than 50 observations in cities $> 500k$ *lll*

All variables including controls are listed in appendix and there are also shown distributions of all the variables. We are controlling for age, age², male, married, employment type dummies (full/part time, self-employed, retired, housewife, student, unemployed, and other), financial satisfaction, health, kids, and importance of religion.

Per controls we follow Pitlik (2016) except that we use financial satisfaction in most specifications instead of income as it arguably predicts freedom more than income and indeed in our sample correlates higher, and also we don't use their country level democracy index, and use PCGDP in appendix only. Likewise, we have all person level predictors that Nikolaev and Bennett (2016) have used.

Money is important for freedom. Indeed as many financial gurus on YouTube would tell you freedom is the only or main reason to accumulate wealth. For instance, the FIRE movement (Financial Independence, Retire Early) is all about freedom/autonomy. Hence in our analyses we will focus on income, class and financial satisfaction. Financial satisfaction is the key—not everyone needs the same amount of money to be financially satisfied and feel free. It is measured as “How satisfied are you with the financial situation of your household?”

We were tempted to add trust, crime victimization and fear of crime, as these correlate with urbanicity and may predict freedom. But eventually we do not include them because those are rather caused by urbanicity, not the other way round, and hence may lead to overcontrol bias:

What we really need, to identify/estimate $X \rightarrow Y$, is to control for other determinants of Y that are also antecedents of X (so, $W \rightarrow X$). If we include controls where the relationship goes the other way ($X \rightarrow W$), we will exacerbate bias in our estimate of $X \rightarrow Y$. (The relevant term for that situation is 'overcontrol bias') (Bartram et al. 2024, p. 5).

5 Results

A natural first approach would be to find overall pattern on pooled complete dataset across all countries to see if there are universal patterns or at least by income group or region. Indeed this is how we have started, but we postpone these results to the appendix as we have learned through our sensitivity/robustness checks that inclusion of country dummies yields all urban-rural results insignificant. Hence we simply start with the results by country.

Financial satisfaction is the strongest correlate of freedom (0.33), stronger than health (0.21), class (0.14), and income (0.09). Financial satisfaction is also related to class (0.26) and income (0.34).

5.1 By country

Results are in figure 2. Is there less freedom in smaller areas as often argued? Are cities more free? No. Most countries show insignificant urban-rural differences at < 0.05 level of significance. Again, it is already a finding worth reporting that people don't have more freedom/autonomy in cities, because there are theories, reasons, and arguments that they do.

*lll*compare the 2 and retain only fig-regCC1mod2

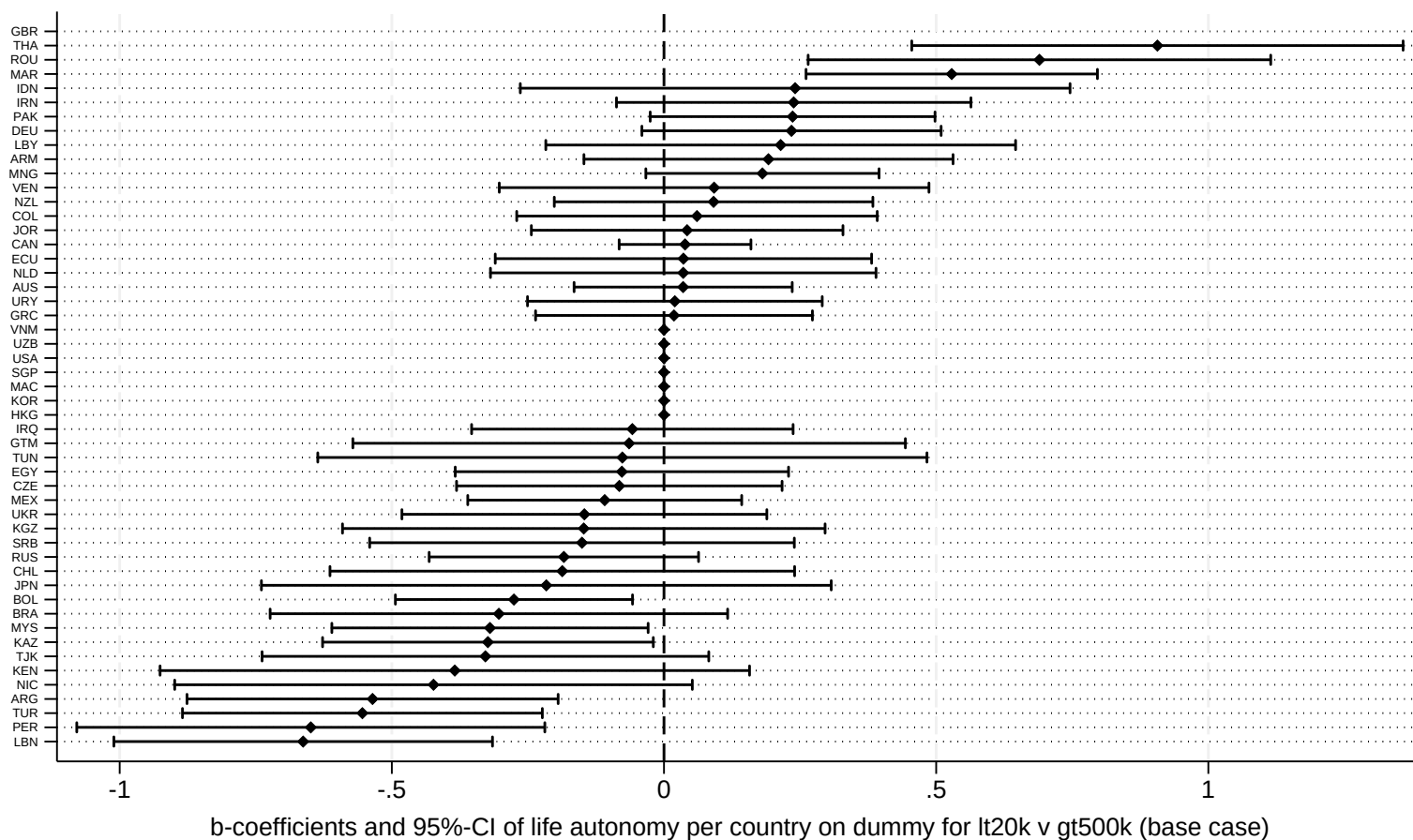
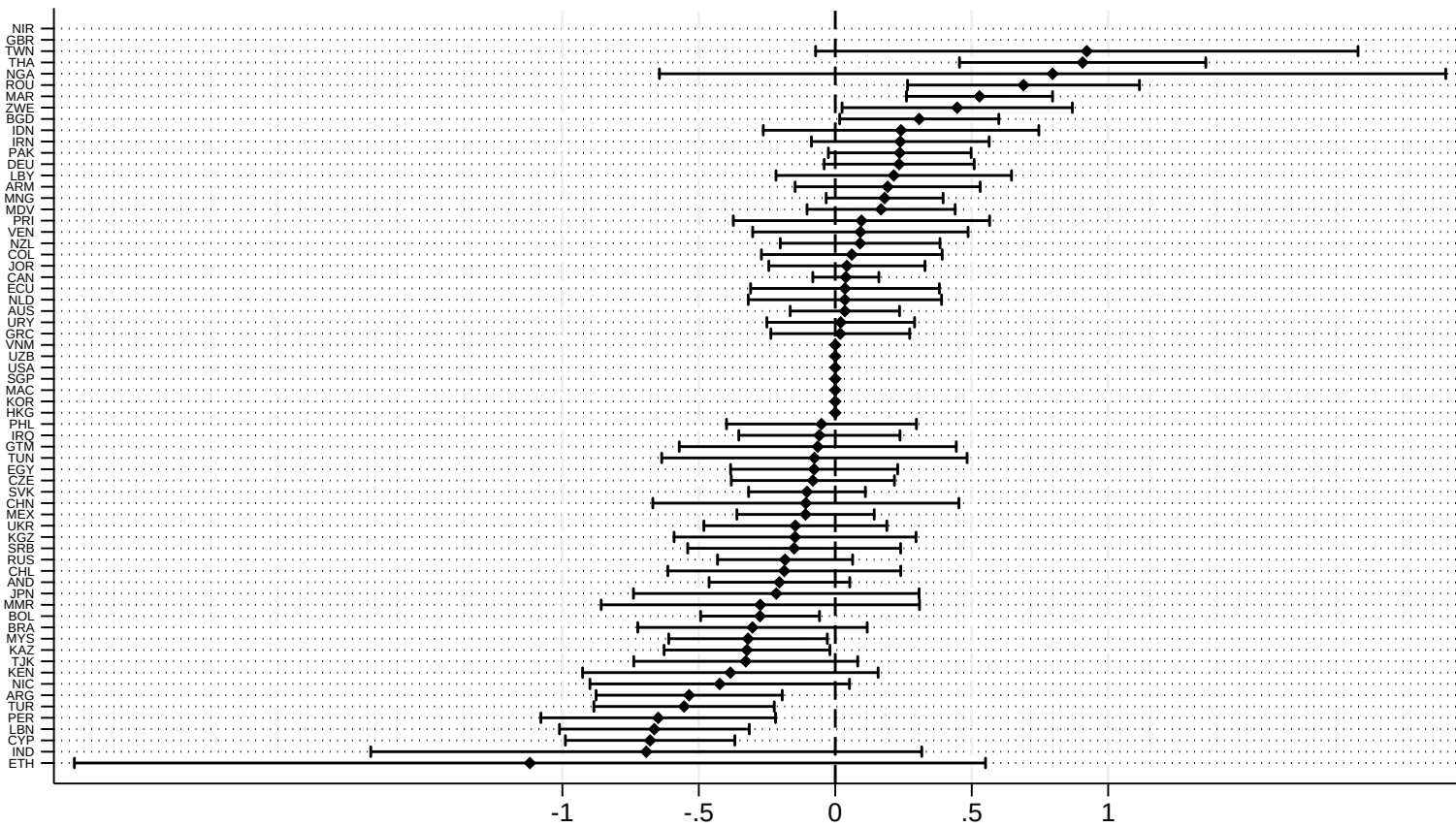


Figure 1: cc goes down from 66 to 51 when dropped if gt500k missing; same pattern as earlier most insignificant, some positive but most negative



b-coefficients and 95%-CI of life autonomy per country on dummy for lt20k v gt500k (base case)

Figure 2: Using model controlling for age, age², male, married, employment type dummies(full/part time, self-employed, retired, housewife, student, unemployed, and other), financial satisfaction, health, kids, and religion important. See appendix for results in table also including omitted here 20 – 500k category. See appendix model for pooled sample in table 3 except that original 8 step urbanicity variable is now categorized into 3 bins as there are smaller cell sizes separately by country. Still note that few countries lack the estimates as even this simplified 3 step urbanicity spectrum is not available. Overall the results are mostly insignificant, but more negative (11) than positive (6). So in general within countries for few significant relationships there is less freedom in smaller areas as theorized in the literature.

Only 16 out of 64 countries have more freedom in cities. They have at least one coefficient on either $< 20k$ or $20 - 500k$ ($v > 500k$) negative and significant at < 0.05 . 6 countries, on the other hand, have more freedom in ruralities—they have positive effect coefficients on smaller areas dummies. Note that figure 2 only shows $< 20k$ coefficients; for $20 - 500k$ coefficients see appendix.

It is difficult to see any consistent country patterns in the table. Perhaps what stands out is that among the richer or more economically developed countries there are smaller differences. In poorer/less economically developed countries there are more differences: mostly less freedom in smaller places.⁷ This makes sense—smaller places in developing countries can be very traditional and very poor—freedom is difficult in such places. In richer countries, on the other hand, one can already be free just about anywhere, no need for a large city. This is similar to what happiness literature have found with respect to happiness (Okulicz-Kozaryn 2015)—only in the very poorest places of all such as Sub-Saharan Africa happiness is greater in larger cities as life is unbearable and lacking necessities such as shelter and clean water in ruralities.

Countries could be classified by region (see appendix). About half of Latin countries in this sample have lower freedom in smaller areas: ARG, BOL, CHL, NIC, and PER; and about the other half are insignificant: BRA, COL, ECU, GTM, PRI, URY, and VEN.

South Asia does not have a clear pattern: BGD is positive and PAK marginally positive, but IND highly negative, and MDV is insignificant.

⁷But sometimes more.

Europe/Central Asia: mostly insignificant: ARM, CZE, GRC, KGZ, NLD, RUS, SRB, SVK, and UKR. But otherwise mostly negative: AND marginally, CYP, KAZ, TJK marginally, and TUR; and positive: ROU, DEU marginally. Note that GBR has missing obs and does not appear in the table at all, and UZB has missing urbanicity data.

East Asia/Pacific is mostly insignificant: AUS, CHN, IDN, MMR, NZL, PHL. And several countries in East Asia/Pacific are without respondents in smaller areas: HKG, KOR, MAC, SGP, and VNM. Positive: MNG, and highly positive THA; and negative: JPN, MYS.

Middle East/North Africa mostly insignificant: EGY, IRN, JOR, LBY, and TUN. Two negative: IRQ, LBN; and one positive: MAR.

Sub-Saharan Africa: two insignificant: KEN, NGA; ETH is negative, but ZWE positive.

Two countries missing the region classification by the World Bank: NIR and TWN. NIR has a small sample size of less than 500 and missing observations on the variables used in regression so it does not appear in the table. TWN is insignificant.

5.2 Interactions with financial satisfaction, class, and income

Again the initial results with pooled sample for all the countries (in appendix) turned out to be all insignificant when including country dummies, hence we started with results within each country in the previous table.

But there is one significant result for the pooled sample on urbanicity when controlling for countries—when urbanicity is interacted with financial satisfaction, class, and income. While earlier we just used OLS within each country, here we use multilevel modeling (MLM) with random intercept at country level.

Results are visualized in figure 3.⁸

In panel a by financial satisfaction those in top 3 categories are at about 8 v middle 3 at slightly above 7 v bottom 4 at below 6.5. For the bottom categories there is very slight decrease from rural to urban, and for the middle and upper financial satisfaction categories there is a slight increase from rural to urban. In panel b class: Upper middle and upper classes are more free in largest cities, while lower class is more free in smallest places. In other words, freedom differences are magnified in largest cities for both financial satisfaction and class—the financially satisfied and higher classes enjoy higher freedom premium over financially dissatisfied and lower class in cities.

Income in panel c agrees for the bottom 2 quintiles where freedom declines from $< 10k$ to $> 500k$, and opposite pattern for 3rd middle quintile and 4th quintile where freedom slightly increases from $< 10k$ to $> 500k$. But the top quintile, counterintuitively, behaves like the bottom 2, and we do not have an explanation for this.

⁸We focus here on visual interpretations using marginsplot, but underlying regressions are in appendix. The relevant significant interactions are for smallest places $< 10k$ (against the base case of $> 500k$ with top category(ies) for financial satisfaction in first column a for middle financial satisfaction (v bottom base case) at -0.12 and the top category at -0.13. Likewise, in column b for class interactions, there is significant interaction of $< 10k * Uppmid/upper$ at -0.16. Thus, folks of both high financial satisfaction and class are less free in smallest areas v the largest areas. In last column c with income, there are no significant interactions except unexpected top income quintile, but other insignificant patterns are in line with financial satisfaction and class: the two bottom deciles are less free in the largest places.

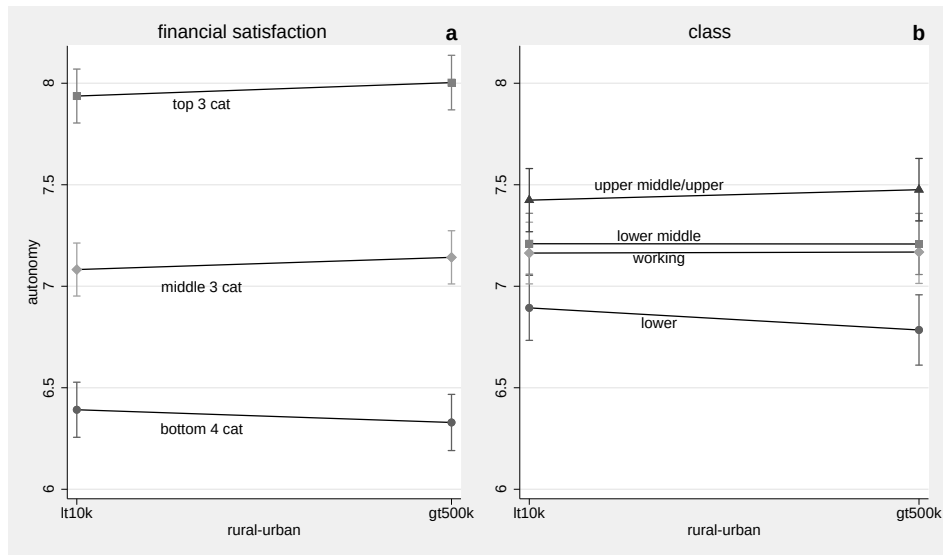


Figure 3: Multilevel country random intercept regressions of freedom on rural-urban dummies interacted with: a) financial satisfaction, and b) class. For underlying regressions see appendix. Controls are the same as in the previous section.

It could be interesting to separate these results by country, however we leave it for the future research as sample sizes run small for interactions. But we do it by region and income group for financial satisfaction only and postpone these results to appendix⁹

6 Discussion and Conclusion

We find support for Toennies' idea: "in the city and therefore, where general conditions characteristic of the Gesellschaft prevail, only the upper strata, the rich and the cultured, are really active and alive" (Tönnies [1887] 2002, p. 227). There are significant interactions in the pooled sample for financial satisfaction and class.

In the pooled sample without interactions there are no significant urban rural differences when controlling for country dummies. But there are some significant results separately by country. The hypothesis that cities are more free does not find support except in 16 countries out of 64.

It is worth highlighting that finding no effect in most countries is a finding worth reporting—the expectation from theory and literature is that the bigger places have more freedom. And in 6 countries we find a positive effect.

Like in Okulicz-Kozaryn (2024) urbanites probably actually have in a way fuller and more free life, but perhaps they realize more that they miss out relatively to other urbanites, and so they feel relatively less free. One mechanism could be that people tend to compare upwards (Frey and Stutzer 2002) and there are more people to compare to in urban areas, inequality ladders are taller and the pond is bigger—small fish in big pond (city) seem to be less free than big fish in a small pond (rural area).

In poorer/less economically developed countries there are more differences in freedom across rural-urban, mostly less freedom in smaller places¹⁰—this makes sense—smaller places in developing countries can be very traditional and very poor—freedom is difficult in such places. In richer countries, on the other hand, one can already be free just about anywhere, no need for a large city. This is a similar finding as in literature with respect to happiness (Okulicz-Kozaryn 2015)—only in the very poorest places of all such as Sub-Saharan Africa, happiness is greater in larger cities as life is unbearable and lacking necessities such as shelter and clean water in

⁹Here just real quick: Financial satisfaction panel a results are stronger in East Asia/Pacific and Sub-Saharan Africa (due to sharp increase for the top tertile), and show similar pattern in Europe/Central Asia except that the middle tertile for financial satisfaction is stronger than the top. Latin America/Caribbean and North America shows the opposite pattern where the bottom tertile has the steepest slope, and in Middle East/North Africa middle and top tertiles even reverse to negative. In South Asia the slopes are parallel—no differences there. By income group: High income is similar to the overall pattern.

¹⁰(but sometimes more)

ruralities. Everywhere else urbanites tend to be less happy (Okulicz-Kozaryn and Valente 2021).

ONLINE APPENDIX (SOM)

[note: this section will NOT be a part of the final version of the manuscript, but will be available online instead as SOM 'Supplementary Online Material']

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1 AI literature review

Gemini query “scholarly literature on individual autonomy or freedom across urban-rural”¹¹ suggested the following literature summarised in this section by us, the human authors.

Autonomy Dimension	Rural	Urban
Social Control	Strict; governed by kinship, mutual surveillance, and traditional norms.	Loose; regulated by formal systems (laws)
Core Values	Interdependence, community solidarity, and generational stability.	Individualism, open-mindedness, and soc
Family Dynamics	High value placed on obedience and material/financial cooperation.	Focus on psychological independence an
Physical Liberty	High geographic isolation; dependency on personal transit for survival.	High physical access via transit; restricte

? is a summary of their themed issue on “Autonomy and independence in urban environments.” Rather suburban than rural contrasts for urban, and more on culture and life cycle than space.

As the world continues to urbanize, ruralities are disappearing, and so there is also “the perceived disappearance of rural childhood and the values associated with it” in Czech republic (?). Rurality in the developed world is increasingly a periphery, loosing power and voice v urban centrality, being abndoned and disrespected—there is rural disconten, resentment, and shame. See also ???. This would

¹¹https://www.google.com/search?q=scholarly+literature+on+individual+autonomy+or+freedom+across+urban-rural&rlz=1CAHWSA_enUS1125US1125&oq=scholarly+literature+on+individual+autonomy+or+freedom+across+urban-rural&gs_lcrp=EgZjaHJvbWUyBggAEEUYOdIBCTIyNDYyajBqN6gCCLACaFEFAI5zaHyURfbxBQC0c2h81EX2&sourceid=chrome&source=chrome.ob&ie=UTF-8

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suggest that ruralities in developed countries may have less autonomy—but that's rather because ruralities have been abandoned and disrespected, not because they are inherently less autonomous. And ruralities became digitized—children do not play outside in natura that much anymore, but instead on screens just like their urban counterparts. And autonomy is lost to screens—kids are dependent or even addicted to them. In general, while there has been a loss of power and respect v urban, not necessarily the autonomy, ruralities still have a good deal of agency.

2 Additional info

2.1 Country sample sizes; Regional country classification; Variables' definitions (survey questions); Histograms

We use World Bank regions classification into several regions: East Asia/Pacific, Europe/Central Asia, Latin America/Caribbean, Middle East/North Africa, North America, South Asia, and Sub-Saharan Africa. Countries in each region and sample sizes are in appendix.

```
. bys region: ta cc
-----
-> regionname =
ISO 3166-1 |
alpha-3 |
country |
code |
-----
Freq.      Percent      Cum.
-----
NIR |      447      26.77      26.77
TWN |     1,223      73.23     100.00
-----
Total |     1,670     100.00
-----
-> regionname = East Asia/Pacific
ISO 3166-1 |
alpha-3 |
country |
code |
-----
Freq.      Percent      Cum.
-----
AUS |     1,813       7.29       7.29
CHN |     3,036     12.21     19.50
HKG |     2,075       8.35     27.85
IDN |     3,200     12.87     40.72
JPN |     1,353       5.44     46.16
KOR |     1,245       5.01     51.16
MAC |     1,023       4.11     55.28
MMR |     1,200       4.83     60.10
MNG |     1,638       6.59     66.69
MYS |     1,313       5.28     71.97
NZL |     1,057       4.25     76.22
PHL |     1,200       4.83     81.05
SGP |     2,012       8.09     89.14
THA |     1,500       6.03     95.17
VNM |     1,200       4.83     100.00
-----
Total |    24,865     100.00
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-> regionname = Europe/Central Asia
ISO 3166-1 |
alpha-3 |
country |
code |
-----
Freq.      Percent      Cum.
-----
AND |     1,004       3.88       3.88
ARM |     1,223       4.73       8.61
CYP |     1,000       3.87     12.48
CZE |     1,200       4.64     17.12
DEU |     1,528       5.91     23.03
GBR |     2,609     10.09     33.13
GRC |     1,200       4.64     37.77
KAZ |     1,276       4.94     42.70
KGZ |     1,200       4.64     47.35
NLD |     2,145       8.30     55.64
ROU |     1,257       4.86     60.51
RUS |     1,810       7.00     67.51
SRB |     1,046       4.05     71.55
SVK |     1,200       4.64     76.20
TJK |     1,200       4.64     80.84
TUR |     2,415       9.34     90.18
UKR |     1,289       4.99     95.16
UZB |     1,250       4.84     100.00
-----
Total |    25,852     100.00
-----
-> regionname = Latin America/Caribbean
ISO 3166-1 |
alpha-3 |
country |
code |
-----
Freq.      Percent      Cum.
-----
ARG |     1,003       5.75       5.75
BOL |     2,067     11.85     17.60
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BRA	1,762	10.10	27.71
CHL	1,000	5.73	33.44
COL	1,520	8.72	42.16
ECU	1,200	6.88	49.04
GTM	1,229	7.05	56.09
MEX	1,741	9.98	66.07
NIC	1,200	6.88	72.95
PER	1,400	8.03	80.98
PRI	1,127	6.46	87.44
URY	1,000	5.73	93.18
VEN	1,190	6.82	100.00

Total	17,439	100.00
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-> regionname = Middle East/North Africa

ISO 3166-1			
alpha-3			
country			
code	Freq.	Percent	Cum.
EGY	1,200	12.11	12.11
IRN	1,499	15.13	27.25
IRQ	1,200	12.11	39.36
JOR	1,203	12.14	51.50
LBN	1,200	12.11	63.62
LBY	1,196	12.07	75.69
MAR	1,200	12.11	87.81
TUN	1,208	12.19	100.00
Total	9,906	100.00	

-> regionname = North America

ISO 3166-1			
alpha-3			
country			
code	Freq.	Percent	Cum.
CAN	4,018	60.75	60.75
USA	2,596	39.25	100.00
Total	6,614	100.00	

-> regionname = South Asia

ISO 3166-1			
alpha-3			
country			
code	Freq.	Percent	Cum.
BGD	1,200	20.25	20.25
IND	1,692	28.55	48.80
MDV	1,039	17.53	66.33
PAK	1,995	33.67	100.00
Total	5,926	100.00	

-> regionname = Sub-Saharan Africa

ISO 3166-1			
alpha-3			
country			
code	Freq.	Percent	Cum.
ETH	1,230	24.86	24.86
KEN	1,266	25.59	50.44
NGA	1,237	25.00	75.44
ZWE	1,215	24.56	100.00
Total	4,948	100.00	

Table 1: Variable definitions.

name	description
freedom	"Some people feel they have completely free choice and control over their lives, while other people feel that what they do has no real effect on what happens to them. Please use this scale where 1 means 'none at all' and 10 means 'a great deal' to indicate how much freedom of choice and control you feel you have over the way your life turns out."
place size	"OBSERVATIONS BY THE INTERVIEWER; Code size of town where interview was conducted"
age	age
male	male
married or living together as married	"Are you currently(READ OUT AND CODE ONE ONLY) 1 'Married' 2 'Living together as married' 3 'Divorced' 4 'Separated' 5 'Widowed' 6 'Single/Never married' 7 'Divorced, Separated or Widow' 8 'Living apart but steady relation (married,cohabitation)'"
Employment status	"Are you employed now or not? IF YES: About how many hours a week? If more than one job: only for the main job 1 'Full time' 2 'Part time' 3 'Self employed' 4 'Retired' 5 'Housewife' 6 'Students' 7 'Unemployed' 8 'Other'"
financial satisfaction	"How satisfied are you with the financial situation of your household?"
class	"Social class (subjective)"
income	"Scale of incomes"
health	"State of health (subjective)"
religion important	"WVS2000: For each of the following aspects, indicate how important it is in your life. Would you say it is: EVS1999: Please say, for each of the following, how important it is in your life. Religion"

Per distributions it is worth noting left-skewed distribution on freedom, unexpectedly most people feel maximally free at 10 out of 1-10, likewise but as expected with urbanism—most people are in the top category $> 500k$, and again left-skewed on religion, probably not unexpected—religiosity is declining, but still strong.

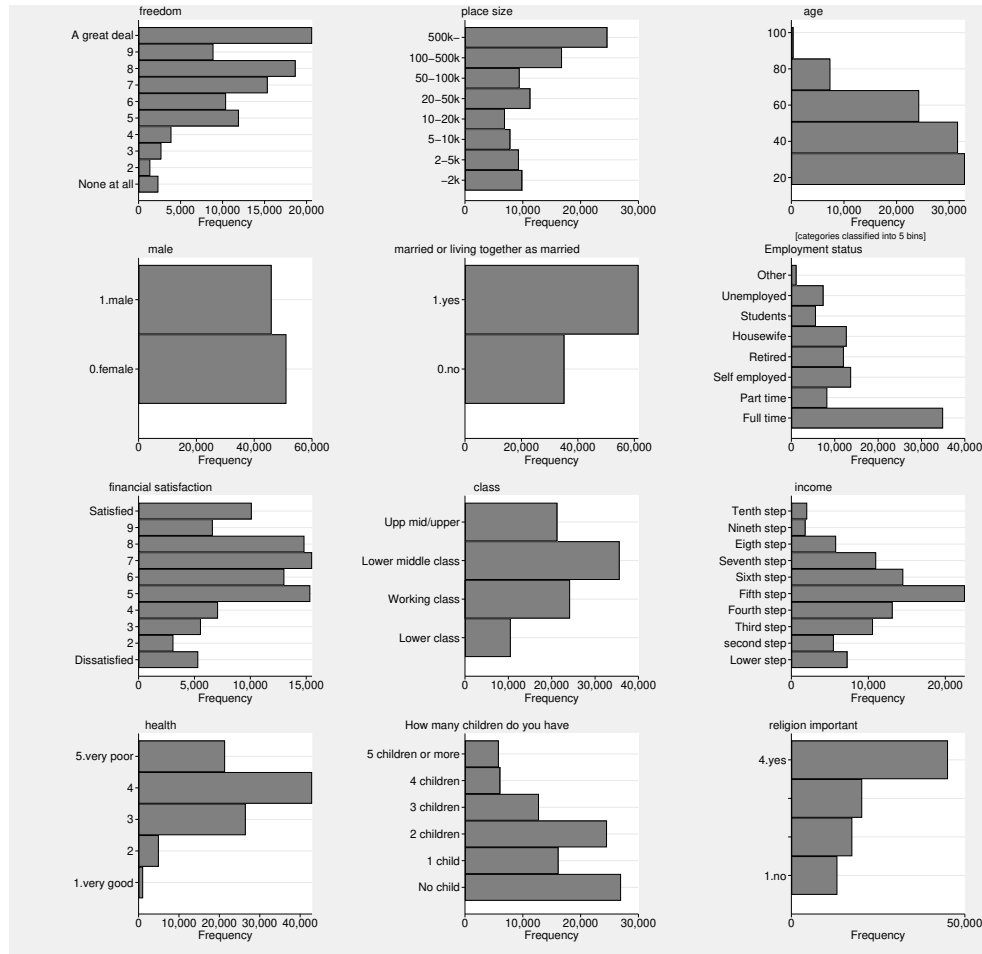


Figure 4: Variables' distribution.

2.2 First/initial results

2.2.1 by region

These were our first/initial results but they do not hold when including country dummies and so we postpone them to appendix.

First for all regions in general there is more freedom in smaller areas v top bin, the base case ($> 500k$) in table 2 but controlling for other predictors in table 3 it is just the 2nd (2-5k) and 3rd (5-10k) smallest categories, and smaller values in 6th and 7th categories. But the very smallest areas ($< 2k$) are not more free. Next we separate by region.

East Asia/Pacific stands out with high effect sizes at about .4 or even .5, also controlling for other predictors of freedom. Europe is mostly less free in smaller areas. Latin America/Caribbean stands out with negative effect in the very smallest areas. Middle East/North Africa has mostly negative effects. North America is positive in 2nd largest category (100-500k). South Asia and Sub-Saharan Africa mostly do not show significant differences.¹²

¹²Except .3 for South Asia for the smallest places ($< 2k$).

	All region dummies	(with dummies)	East Asia/Pacific	Europe/Central Asia	Latin America/Caribbean	Middle East/North Africa	North America	South Asia	Sub-Saharan Africa
town==2k	-0.05+		0.05	-0.13*	-0.28***	0.22	-0.07	0.01	-0.29
town==2-5k	0.14***		0.36***	0.09	-0.13	-0.09	-0.07	0.02	0.06
town==5-10k	0.20***		0.51***	0.03	-0.01	-0.12	0.13	0.03	0.35
town==10-20k	0.09**		0.52***	-0.15**	0.07	-0.34***	-0.03	-0.02	0.19
town==20-50k	0.12***		0.28***	0.08+	0.08	-0.17*	0.09	0.03	-0.22
town==50-100k	0.12***		0.41***	-0.10+	0.04	-0.02	0.12	-0.41+	-0.06
town==100-500k	0.08***		0.55***	-0.12**	-0.08+	-0.27***	0.18**	0.04	-0.45+
constant	6.94***		6.76***	7.06***	7.85***	6.88***	7.44***	7.52***	7.14***
N	93414		24560	25289	17285	9846	5833	5907	4694

+ 0.10 * 0.05 **
0.01 *** 0.001; robust std err

Table 2: OLS regressions of freedom/agency

	All region dummies	(with dummies)	East Asia/Pacific	Europe/Central Asia	Latin America/Caribbean	Middle East/North Africa	North America	South Asia	Sub-Saharan Africa
town==2k	-0.03		-0.00	-0.03	-0.18**	0.02	-0.06	0.28*	-0.17
town==2-5k	0.09**		0.28***	0.00	-0.14	-0.13	-0.10	0.24+	0.18
town==5-10k	0.11***		0.36***	0.02	-0.03	-0.15+	-0.03	0.13	0.33
town==10-20k	0.03		0.41***	-0.20***	-0.00	-0.33***	-0.00	0.27	0.19
town==20-50k	0.04		0.24***	-0.01	-0.01	-0.22**	0.05	0.02	-0.17
town==50-100k	0.06*		0.34***	-0.21***	-0.01	-0.07	0.13+	0.10	0.02
town==100-500k	0.06**		0.48***	-0.18***	-0.03	-0.30***	0.13*	0.07	-0.43+
age	0.01***		0.01*	0.00	0.02**	0.00	0.03***	0.02	0.02
age2	-0.00***		-0.00**	-0.00	-0.00+	0.00	-0.00**	-0.00	-0.00
male	0.00		-0.09***	0.05	-0.08*	0.23***	-0.13**	0.16+	0.14
married or living together as married	-0.01		-0.00	-0.06+	-0.05	0.01	-0.00	0.06	0.27*
Full time	0.00		0.00	0.00	0.00	0.00	0.00	0.00	0.00
Part time	-0.10***		-0.04	-0.09+	-0.24***	-0.05	0.08	-0.32**	-0.06
Self employed	-0.07**		-0.12**	0.08	-0.04	-0.10	0.32***	-0.47***	0.15
Retired	-0.01		0.15**	-0.08	0.05	0.22*	-0.07	-0.31	-0.15
Housewife	-0.22***		-0.19***	-0.19***	-0.05	-0.17*	-0.05	-0.54***	-0.15
Students	-0.21***		-0.22***	-0.08	-0.27***	-0.21*	0.09	-0.31	-0.12
Unemployed	-0.10**		-0.11+	-0.16*	-0.02	-0.14	-0.14	-0.28+	0.12
Other	-0.23**		-0.29+	-0.34*	0.01	-0.20	-0.03	-1.87***	1.53**
financial satisfaction	0.27***		0.31***	0.31***	0.19***	0.31***	0.26***	0.31***	0.10***
health	0.34***		0.32***	0.25***	0.37***	0.47***	0.51***	0.21***	0.24***
How many children do you have	0.04***		-0.01	0.05***	0.02	0.10***	0.05**	0.03	-0.10*
religion important	0.06***		0.09***	0.01	0.03	0.37***	0.08***	0.17**	0.24*
constant	3.55***		3.23***	4.26***	4.79***	1.76***	2.89***	3.28***	4.26***
N	86653		23321	21600	16670	9427	5776	5231	4628

+ 0.10 * 0.05 **
0.01 *** 0.001; robust std err

Table 3: OLS regressions of freedom/agency

2.2.2 Insignificant results with country dummies

Country dummies render cross-country patterns insignificant.

Here we conduct important sensitivity/robustness checks. First in a1 just urbanicity dummies. Then a1a adds region dummies like earlier in the first table little change. A1b has income level dummies TODO spell out; again little change; but addition of country dummies in a1c kills the significance and turns many categories negative notably the first smallest one, similar with controls in models a2* and country dummies make categories even more negative in a2c than in a1c. And then a1c2 and a2c2 add clustered standard errors on country (stata: robust cluster(country))—all significance disappears except in a2c2 on the smallest areas—hence only the very smallest areas < 2k are less free than the largest ones > 500k.

in table 4 we conduct additional robustness checks, a1 and a2 are repeated from the body; a1a and a2a include region dummies: in a1a 100 – 500k results are attenuated, but others are stronger; and in a2a the top three categories spanning 20 – 500k are weaker but other categories are stronger and the smaller category < 2k positive effect goes away. But then in a1c and a2c addition of country dummies kills the effects unfortunately. And having clustered standard errors yields all results insignificant

	a1	a1a	a1b	a1c	a1c2	a2	a2a	a2b	a2c	a2c2
town==2k	-0.03	-0.05+	-0.03	-0.07*	-0.07	-0.06*	-0.03	-0.06*	-0.06+	-0.06
town==2-5k	0.08**	0.14***	0.10**	-0.05	-0.05	0.03	0.09**	0.04	-0.03	-0.03
town==5-10k	0.16***	0.20***	0.16***	0.00	0.00	0.07*	0.11***	0.07*	-0.02	-0.02
town==10-20k	0.03	0.09**	0.02	-0.12***	-0.12	0.00	0.03	-0.01	-0.09**	-0.09
town==20-50k	0.14***	0.12***	0.13***	-0.02	-0.02	0.07**	0.04	0.07**	-0.04	-0.04
town==50-100k	0.13***	0.12***	0.11***	-0.01	-0.01	0.10***	0.06*	0.08**	-0.01	-0.01
town==100-500k	0.13***	0.08***	0.11***	-0.02	-0.02	0.11***	0.06**	0.09***	-0.03	-0.03
age						0.01+	0.01***	0.00	0.02***	0.02***
age2						-0.00*	-0.00***	-0.00+	-0.00***	-0.00***
male						0.02	0.00	0.02	0.01	0.01
married or living together as married						-0.08***	-0.01	-0.07***	0.00	0.00
Full time						0.00	0.00	0.00	0.00	0.00
Part time						-0.09***	-0.10***	-0.09**	-0.09***	-0.09*
Self employed						0.02	-0.07**	0.02	-0.02	-0.02
Retired						0.02	-0.01	0.01	-0.05+	-0.05
Housewife						-0.16***	-0.22***	-0.16***	-0.20***	-0.20***
Students						-0.19***	-0.21***	-0.19***	-0.14***	-0.14**
Unemployed						-0.01	-0.10**	-0.00	-0.11***	-0.11*
Other						-0.07	-0.23**	-0.04	-0.25***	-0.25*
financial satisfaction						0.27***	0.27***	0.27***	0.26***	0.26***
health						0.35***	0.34***	0.34***	0.34***	0.34***
How many children do you have						0.07***	0.04***	0.07***	0.02*	0.02
religion important						0.07***	0.06***	0.07***	0.03***	0.03*
region dummies	no	yes	no	no	no	no	yes	no	no	no
country dummies	no	no	no	yes	yes	no	no	no	yes	yes
income level dummies	no	no	yes	no	no	no	no	yes	no	no
constant	7.14***	6.94***	7.14***	7.57***	7.57***	3.84***	3.55***	3.87***	3.99***	3.99***
N	95083	93414	93414	95083	95083	87874	86653	86653	87874	87874

+ 0.10 * 0.05 **
0.01 *** 0.001; robust std err

Table 4: OLS regressions of freedom/agency

2.3 Main results from paper

2.3.1 By country

First let's start with sample sizes by country by 3 urbanicity categories—in general they are sizable > 100, and in cases where they are not CI is wider on the estimates such as with ETH at -1.12 on lt20k that is insignificant

ISO 3166-1 alpha-3 country code	RECODE of town (place size)			Total
	lt20k	20-500k	gt500k	
AND	730	274	0	1,004
ARG	200	180	623	1,003
ARM	545	232	446	1,223
AUS	453	567	705	1,725
BGD	991	209	0	1,200
BOL	625	741	701	2,067
BRA	185	864	713	1,762
CAN	1,098	1,694	1,226	4,018
CHL	152	487	361	1,000
CHN	50	2,986	0	3,036
COL	304	722	494	1,520
CYP	615	385	0	1,000
CZE	680	370	150	1,200
DEU	712	616	200	1,528
ECU	384	466	350	1,200
EGY	424	456	320	1,200
ETH	841	380	9	1,230
GBR	495	1,713	397	2,605
GRC	420	340	440	1,200
GTM	71	505	653	1,229
HKG	0	0	2,075	2,075
IDN	2,761	346	93	3,200
IND	1,415	235	42	1,692
IRN	400	609	490	1,499
IRQ	320	232	648	1,200
JOR	375	306	522	1,203
JPN	51	786	516	1,353
KAZ	457	456	363	1,276
KEN	499	421	122	1,042
KGZ	710	290	200	1,200
KOR	0	364	881	1,245
LBN	410	660	130	1,200
LBY	480	544	172	1,196
MAC	0	0	1,023	1,023
MAR	380	480	340	1,200
MDV	738	301	0	1,039

MEX	858	463	403	1,724
MMR	1,116	84	0	1,200
MNG	378	257	1,003	1,638
MYS	744	428	141	1,313
NGA	848	375	14	1,237
NIC	381	709	110	1,200
NIR	39	407	0	446
NLD	104	1,710	217	2,031
NZL	420	387	187	994
PAK	1,272	249	474	1,995
PER	355	945	100	1,400
PHL	910	290	0	1,200
PRI	73	1,054	0	1,127
ROU	703	435	119	1,257
RUS	594	626	590	1,810
SGP	0	0	2,012	2,012
SRB	403	447	196	1,046
SVK	750	450	0	1,200
THA	1,244	116	140	1,500
TJK	880	210	110	1,200
TUN	514	614	80	1,208
TUR	240	1,767	408	2,415
TWN	40	1,165	18	1,223
UKR	638	355	296	1,289
URY	270	290	440	1,000
USA	0	1,443	385	1,828
UZB	0	0	1,250	1,250
VEN	180	661	349	1,190
VNM	0	1,049	151	1,200
ZWE	945	270	0	1,215

Total	33,870	37,473	24,598	95,941

.19

	lt20k	20-500k	N
AND	-0.20+	0.00	997
ARG	-0.54*	-0.34+	849
ARM	0.19	-0.03	1192
AUS	0.04	-0.02	1575
BGD	0.31*	0.00	1197
BOL	-0.28*	-0.25*	1990
BRA	-0.30	-0.03	1687
CAN	0.04	0.08	4018
CHL	-0.19	-0.38*	903
CHN	-0.11	0.00	2775
COL	0.06	0.05	1520
CYP	-0.68*	0.00	943
CZE	-0.08	-0.05	1186
DEU	0.23	0.28+	1491
ECU	0.04	0.02	1184
EGY	-0.08	0.03	1175
ETH	-1.12	-1.43*	1215
GRC	0.02	0.15	1161
GTM	-0.06	-0.01	1113
HKG			2044
IDN	0.24	0.39	3154
IND	-0.69*	-1.12*	1063
IRN	0.24	0.18	1477
IRQ	-0.06	-0.36*	1177
JOR	0.04	0.10	1202
JPN	-0.22	-0.31*	1130
KAZ	-0.32*	-0.15	1116
KEN	-0.38	-0.02	1003
KGZ	-0.15	-0.05	1131
KOR		-0.03	1245
LBN	-0.66*	-0.39*	1200
LBY	0.21	-0.29	923
MAC			792
MAR	0.53*	0.28*	1200
MDV	0.17	0.00	1005
MEX	-0.11	-0.07	1619
MMR	-0.27	0.00	1200
MNG	0.18	0.50*	1638
MYS	-0.32*	-0.17	1313
NGA	0.80	0.16	1212
NIC	-0.42+	-0.20	1200
NLD	0.04	-0.12	1675
NZL	0.09	0.02	832
PAK	0.24+	-0.00	1966
PER	-0.65*	-0.33+	1387
PHL	-0.05	0.00	1199
PRI	0.10	0.00	1065
ROU	0.69*	0.56*	1088
RUS	-0.18	-0.06	1676
SGP			1966
SRB	-0.15	-0.25	980
SVK	-0.10	0.00	1175
THA	0.91*	1.29*	1258
TJK	-0.33+	-0.28	1200
TUN	-0.08	0.14	1073
TUR	-0.55*	-0.39*	2357
TWN	0.92	0.89	1221
UKR	-0.15	0.09	1116
URY	0.02	0.23	966
USA		-0.03	1758
UZB			1116
VEN	0.09	-0.18	1187
VNM		0.22	1200
ZWE	0.45*	0.00	1198
+0.10 *0.05 **0.01 ***0.001; robust std err			

Table 5: OLS regressions of freedom/autonomy on urbanicity dummies (base: $> 500k$) using model controlling for age, age², male, married, employment type dummies(full/part time, self-employed, retired, housewife, student, unemployed, and other), financial satisfaction, health, kids, and religion important. See model for pooled sample in table 3 except that original 8 step urbanicity variable is now categorized into 3 bins as there are smaller cell sizes separately by country. Still note that few countries lack the estimates as even this simplified 3 step urbanicity spectrum is not available. Overall the results are mostly insignificant, but more negative (11) than positive (6). So in general within countries for few significant relationships there is less freedom in smaller areas as theorized in the literature.

Then we split by tertile within a country—the idea is that we will have retained more observations as oppose to forcing $> 500k$ only to be a city, and we do within country urbanicity where local conditions may be different and already a place of 100 – 500k may be large. For instance earlier AND had 0 observations in $> 500k$ category, but here we have 274 obs in 3rd tertile.

We were thinking about median split, but we went with tertiles as there are theoretical reasons to contrast rural places with large cities (Okulicz-Kozaryn 2017, 2015).

We start below with counts by tertile and country.

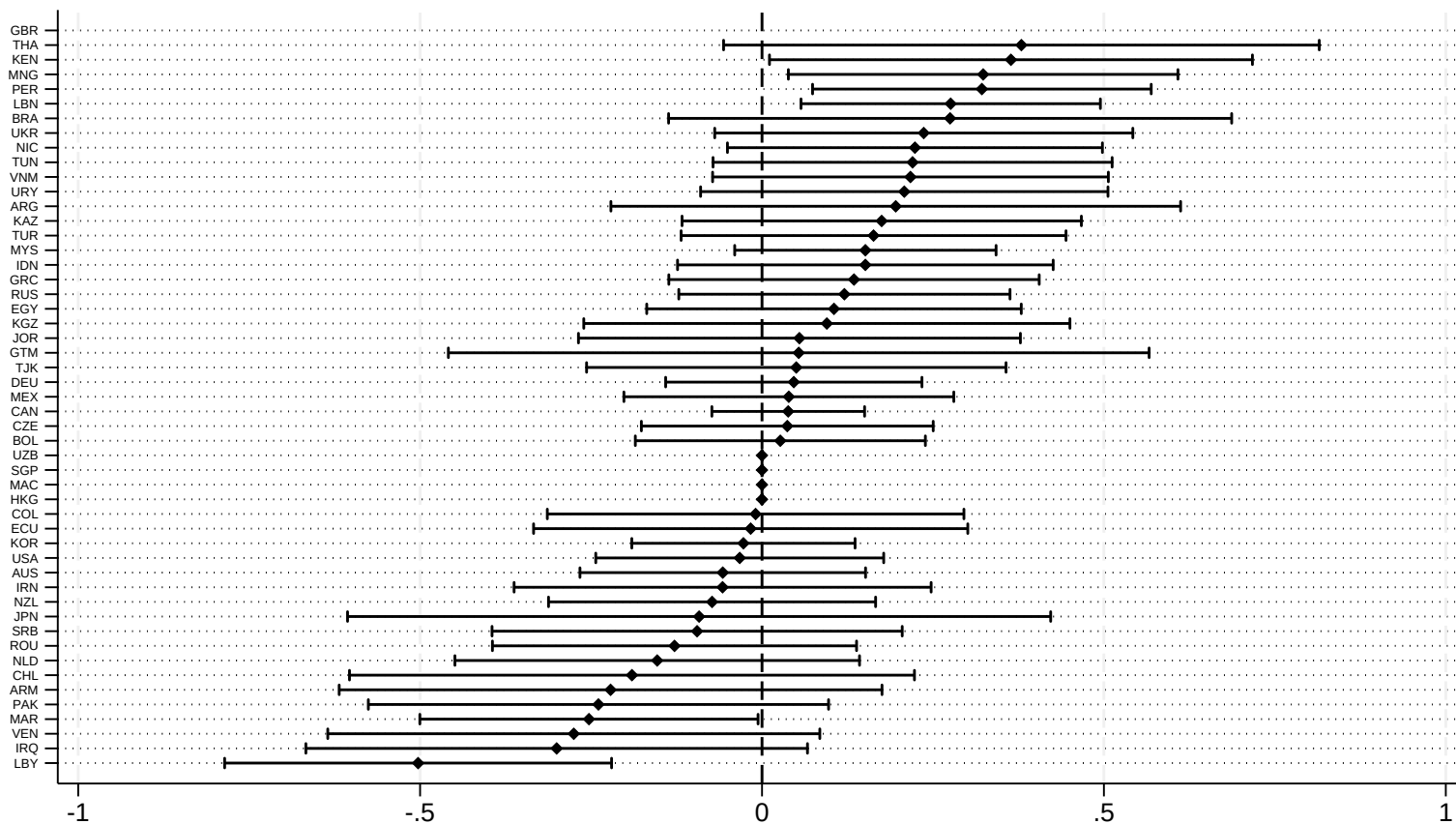
ta cc tertile				
ISO 3166-1				
alpha-3				
country	tertile			
code	1	2	3	Total
-----	-----	-----	-----	-----

AND	501	229	274	1,004
ARG	380	623	0	1,003
ARM	448	775	0	1,223
AUS	625	1,100	0	1,725
BGD	501	335	364	1,200
BOL	879	1,188	0	2,067
BRA	1,049	713	0	1,762
CAN	1,471	1,321	1,226	4,018
CHL	639	361	0	1,000
CHN	1,980	1,056	0	3,036
COL	690	336	494	1,520
CYP	367	333	300	1,000
CZE	460	400	340	1,200
DEU	712	359	457	1,528
ECU	509	341	350	1,200
EGY	424	456	320	1,200
ETH	554	287	389	1,230
GBR	989	1,219	397	2,605
GRC	420	780	0	1,200
GTM	576	653	0	1,229
HKG	2,075	0	0	2,075
IDN	1,442	791	967	3,200
IND	799	389	504	1,692
IRN	599	410	490	1,499
IRQ	552	648	0	1,200
JOR	487	716	0	1,203
JPN	837	516	0	1,353
KAZ	457	456	363	1,276
KEN	395	329	318	1,042
KGZ	450	380	370	1,200
KOR	1,245	0	0	1,245
LBN	410	559	231	1,200
LBY	480	349	367	1,196
MAC	1,023	0	0	1,023
MAR	520	340	340	1,200
MDV	494	200	345	1,039
MEX	707	614	403	1,724
MMR	648	240	312	1,200
MNG	575	1,063	0	1,638
MYS	580	321	412	1,313
NGA	455	393	389	1,237
NIC	720	220	260	1,200
NIR	161	143	142	446
NLD	751	1,063	217	2,031
NZL	420	387	187	994
PAK	896	438	661	1,995
PER	475	825	100	1,400
PHL	610	210	380	1,200
PRI	726	262	139	1,127
ROU	567	343	347	1,257
RUS	756	464	590	1,810
SGP	2,012	0	0	2,012
SRB	365	485	196	1,046
SVK	540	390	270	1,200
THA	923	138	439	1,500
TJK	730	80	390	1,200
TUN	442	495	271	1,208
TUR	2,007	0	408	2,415
TWN	604	601	18	1,223
UKR	517	476	296	1,289
URY	410	590	0	1,000
USA	635	808	385	1,828
UZB	1,250	0	0	1,250
VEN	401	440	349	1,190
VNM	1,049	0	151	1,200
ZWE	768	152	295	1,215
Total	48,139	30,589	17,213	95,941

.19

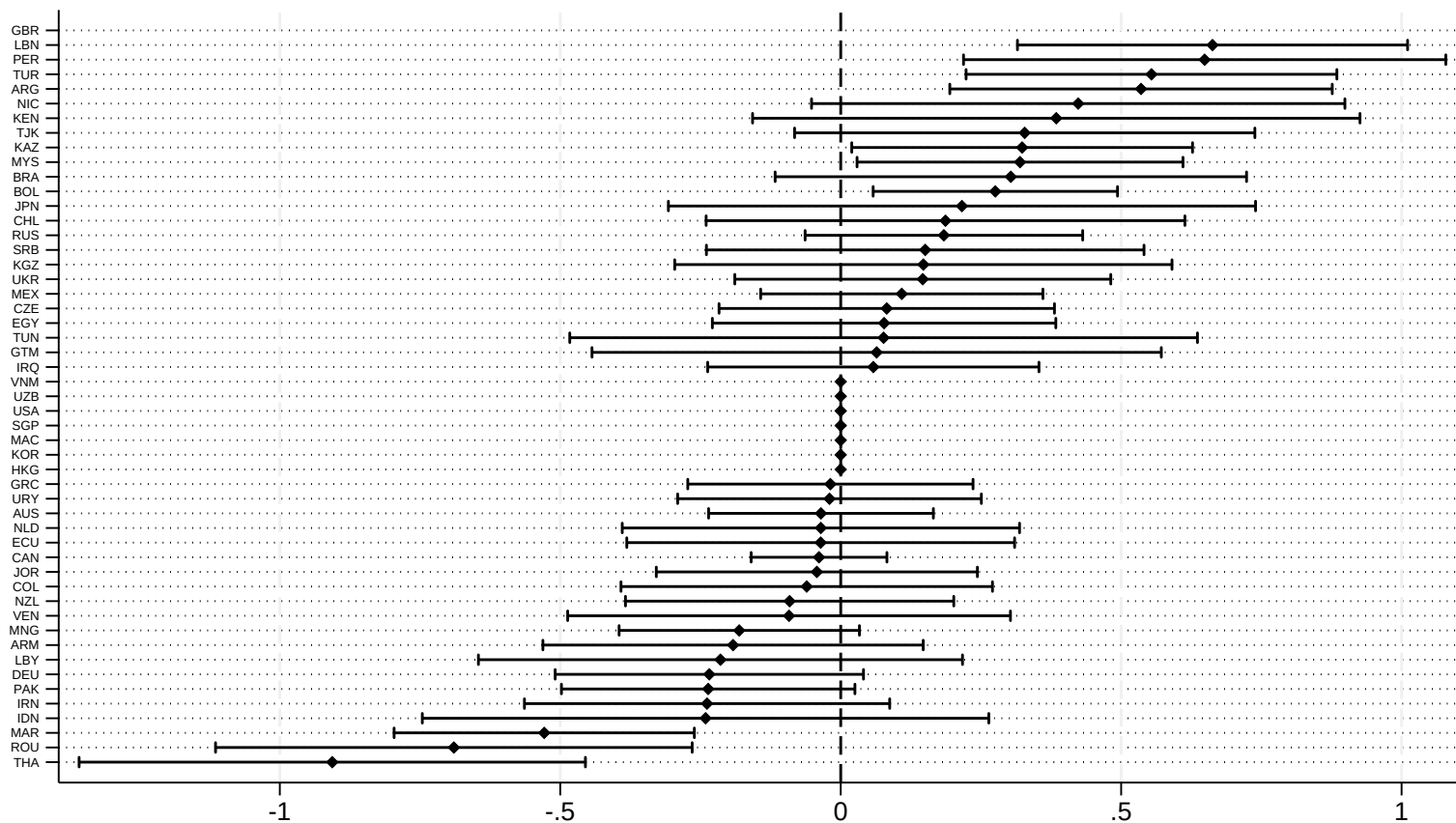
Then we repeat twostep graph from the body just with the 1st category as a base in figure 5 and 6 so we can easier compare it to our tertile estimates.

Overall without controls more freedom in cities as hypothesized, but still most effects are insignificant, and with controls results are even less significant.



b-coefficients and 95%-CI of life autonomy per country on dummy for 100-500k v lt20k (base case); dropped c if cat gt500k mis

Figure 5: fff2



b-coefficients and 95%-CI of life autonomy per country on dummy for 500k v lt20k (base case); dropped c if cat gt500k missing

Figure 6: fff3

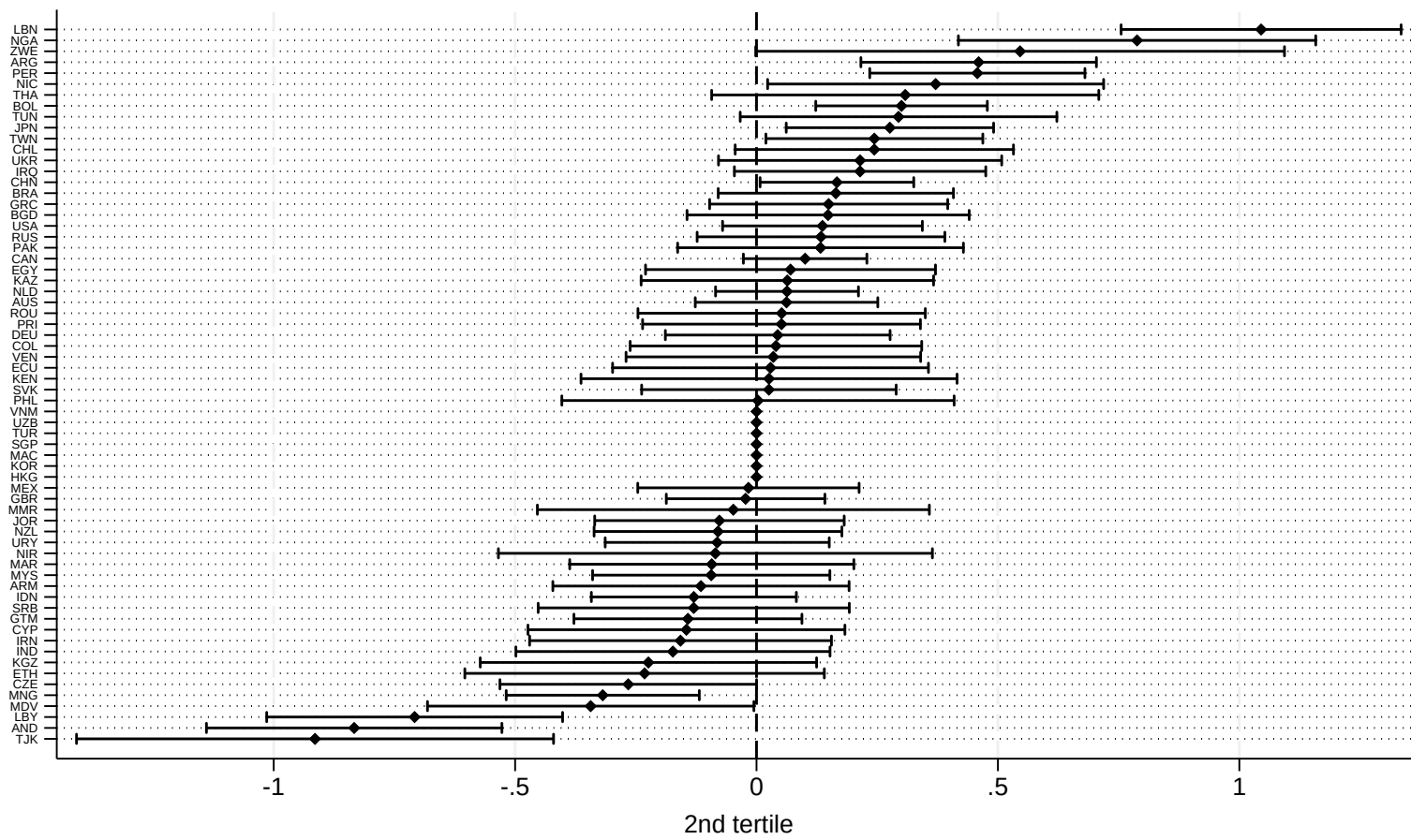


Figure 7: 2nd tertile: just 2 tertile dummies on rhs, 1st tertile base

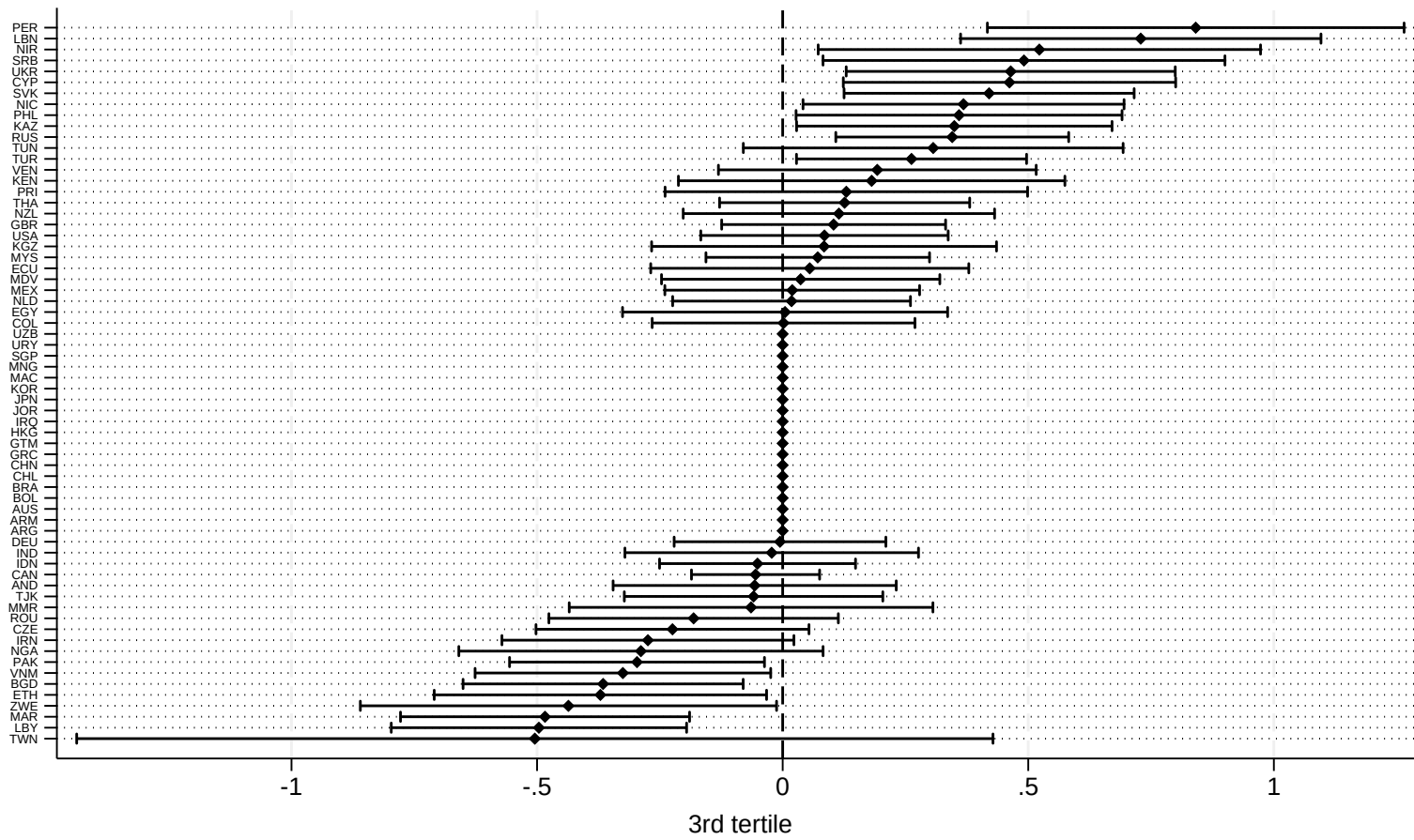


Figure 8: 3rd tertile: just 2 tertile dummies on rhs, 1st tertile base

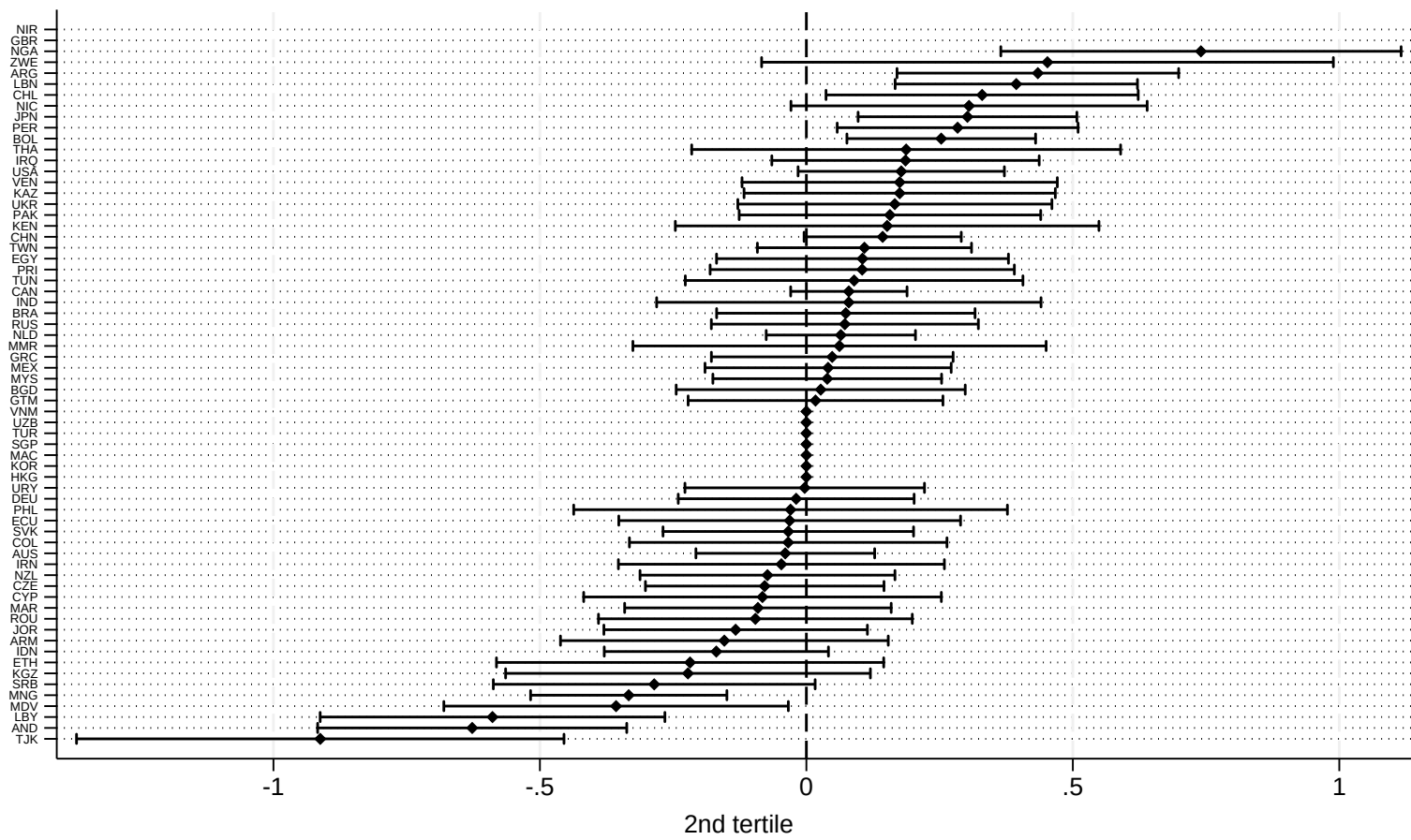


Figure 9: 2nd tertile: all controls

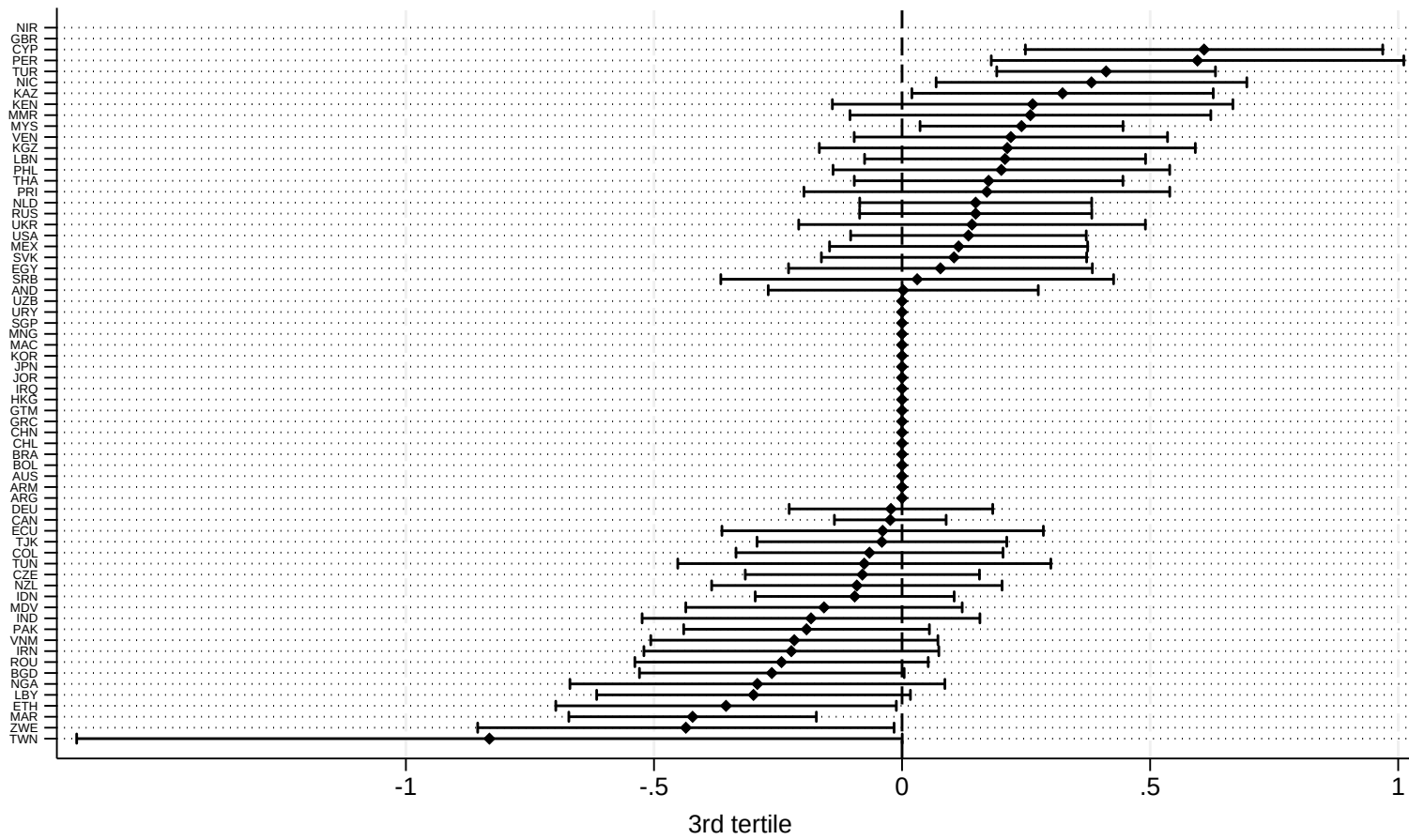


Figure 10: 3rd tertile: all controls

We explore further by interacting region dummies with urban-rural in figure 11. We find significant interactions for Latin America and South Asia (v East Asia/Pacific)—it is difficult to see patterns in figure with all regions—we plot only the significant interactions in figure 12.

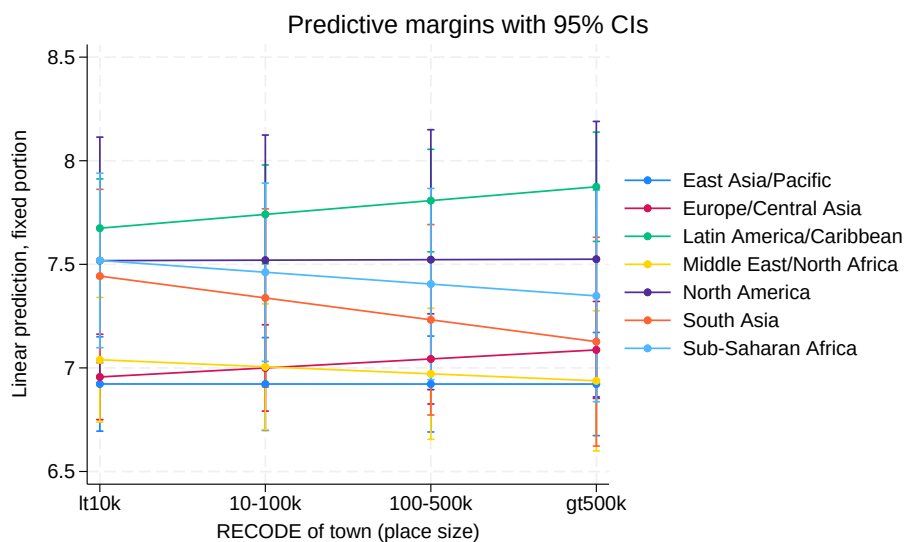


Figure 11: Interactions of region dummies with urban-rural.

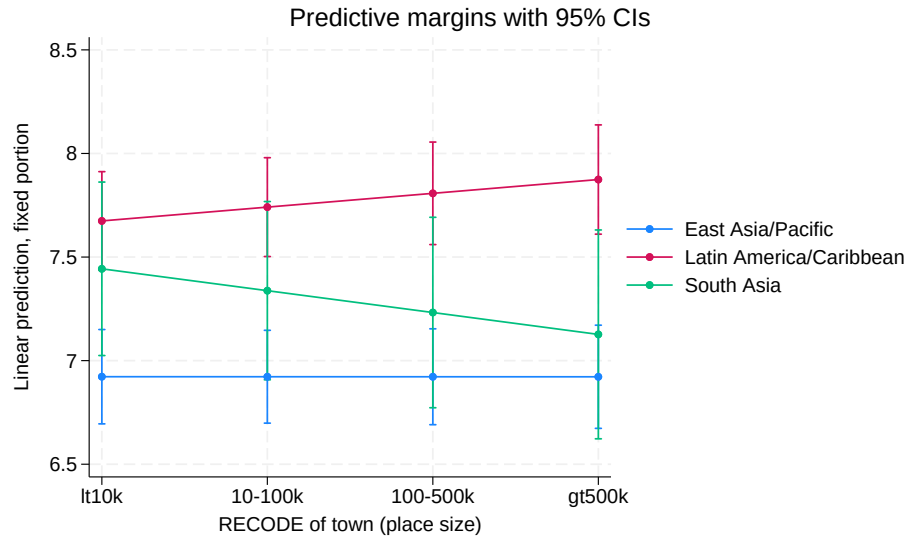


Figure 12: Interactions of region dummies with urban-rural.

Last, we interacted urban-rural at person level with % urban at country level and the interaction is marginally significant at 0.1 level of significance and we plot results in figure 13. Marginsplot shows that in countries that are less urban, there is more freedom in smaller places, but in more urban countries there is more freedom in large cities. It is an intriguing finding and we do not have a good explanation for it. Perhaps in heavily urbanized countries rural barely exists and it does not help much with freedom, while in more rural countries there is more meaningful distinction between urban and rural and there is more freedom in rural areas.

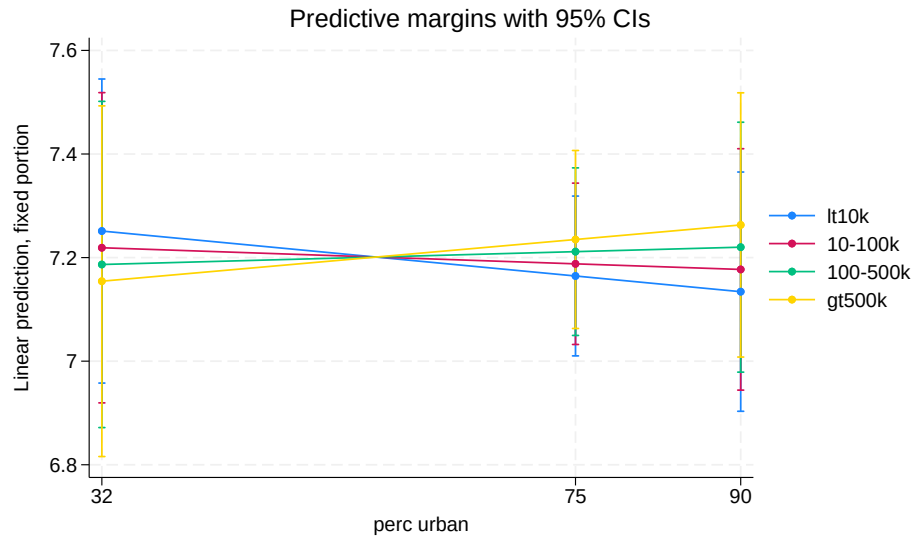


Figure 13: Interaction of urban-rural at person level with % urban at country level. X-axis values are chosen based on the variable's distribution: 31.2 is 5th percentile, median is 75, and 90 is 92nd percentile.

We have used above just the usual World Bank definition, which uses each country's statistical office definition—for discussion see especially Mikhaeil et al. (2024) and also Deb and Okulicz-Kozaryn (2023).

We have also tried similar cross-level interactions with PCGDP and income inequality, but didn't find significant interactions.

Last we check alternative specification if we are not overcontrolling (Bartram et al. 2024)—in the body of the paper we used a set of control variables from the literature, however, we think that we may drop gender, age and age2—as they may not be predictive of

urbanicity. First, we repeat graph from the body for ease of comparison in figure 14. And below it figure without the above mentioned controls 15. Results are substantively the same. And in 16 adding education, eg more educated persons may move to a city—again, very similar results. And in 17 fewer countries but still most insignificant.

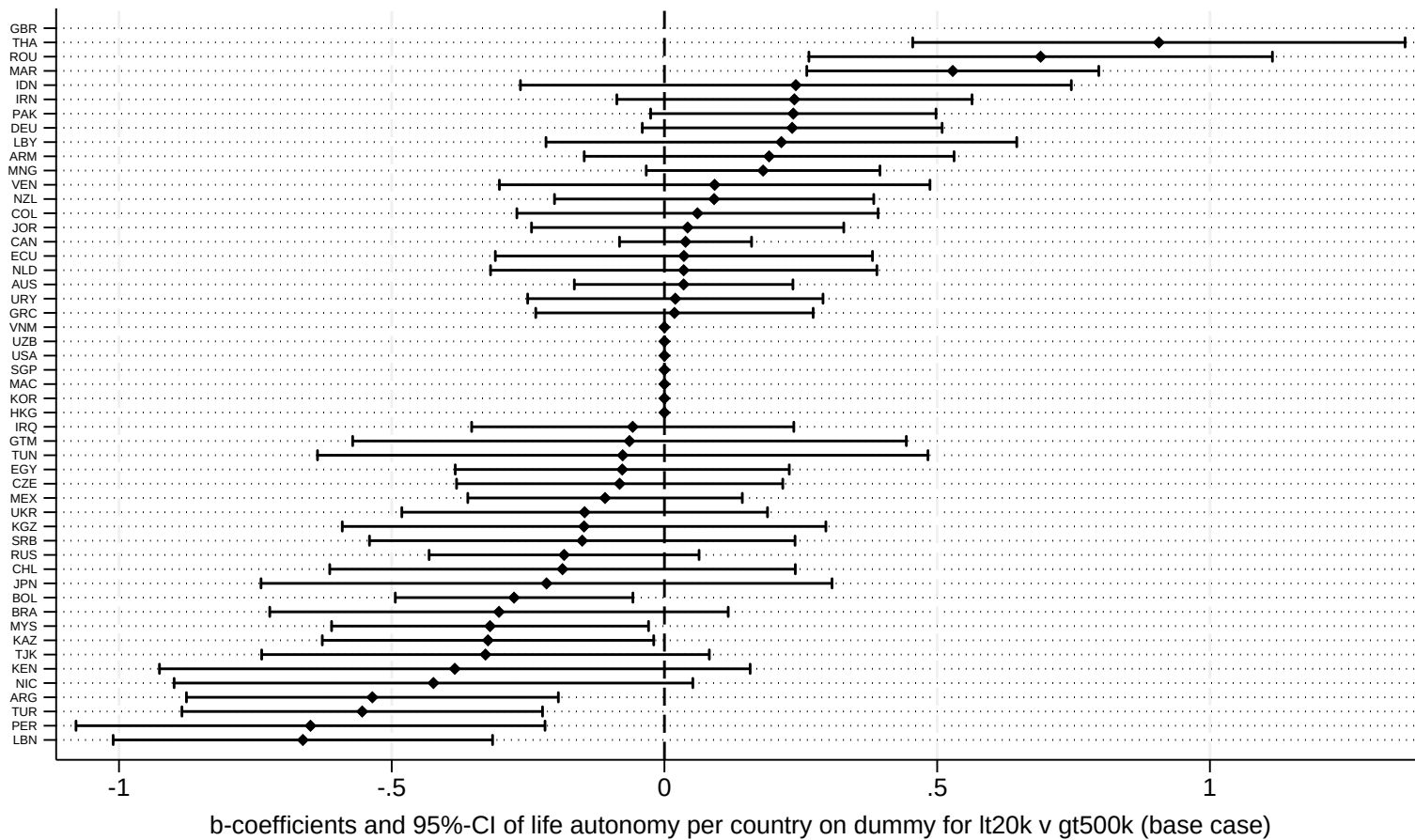
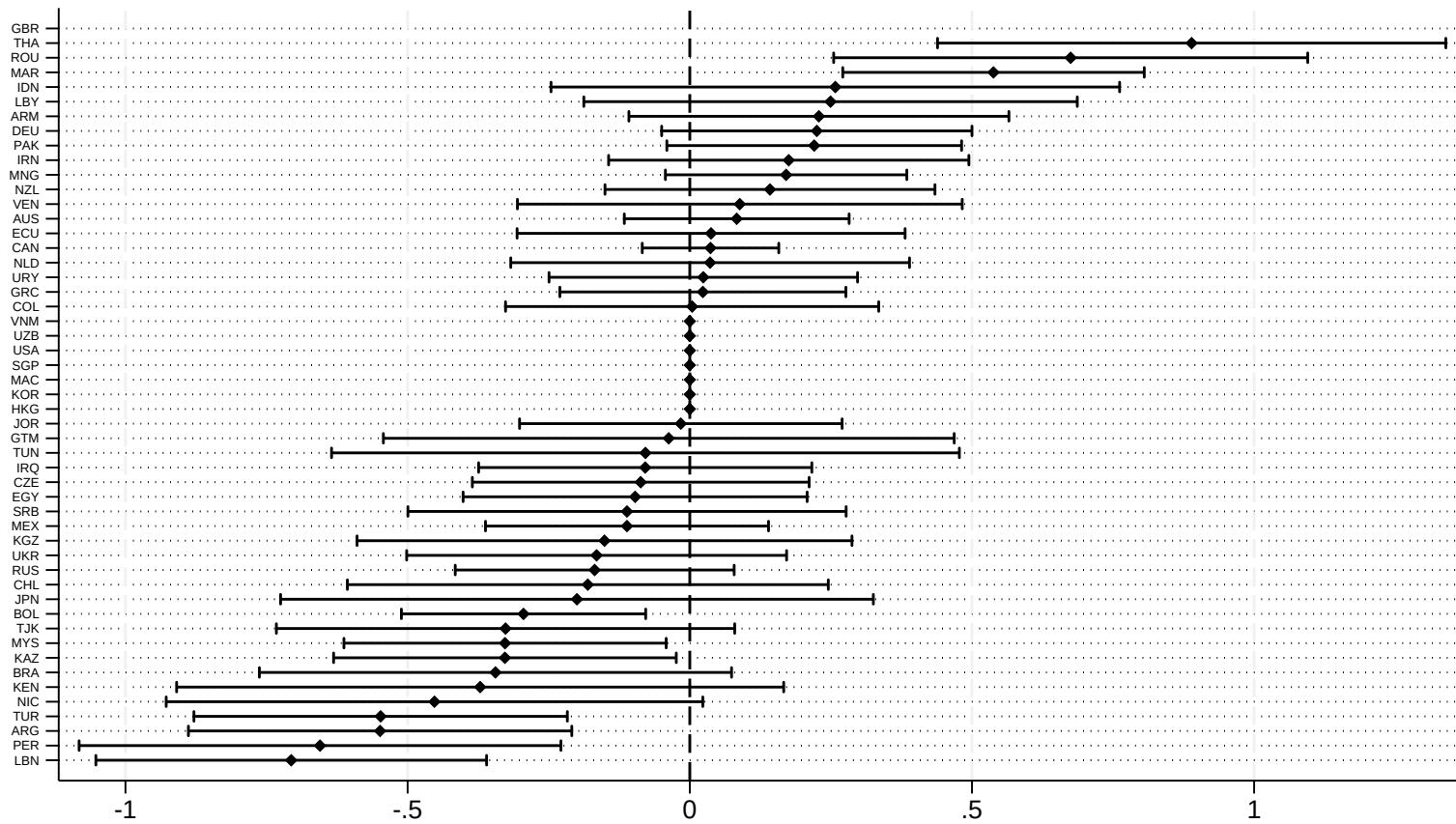
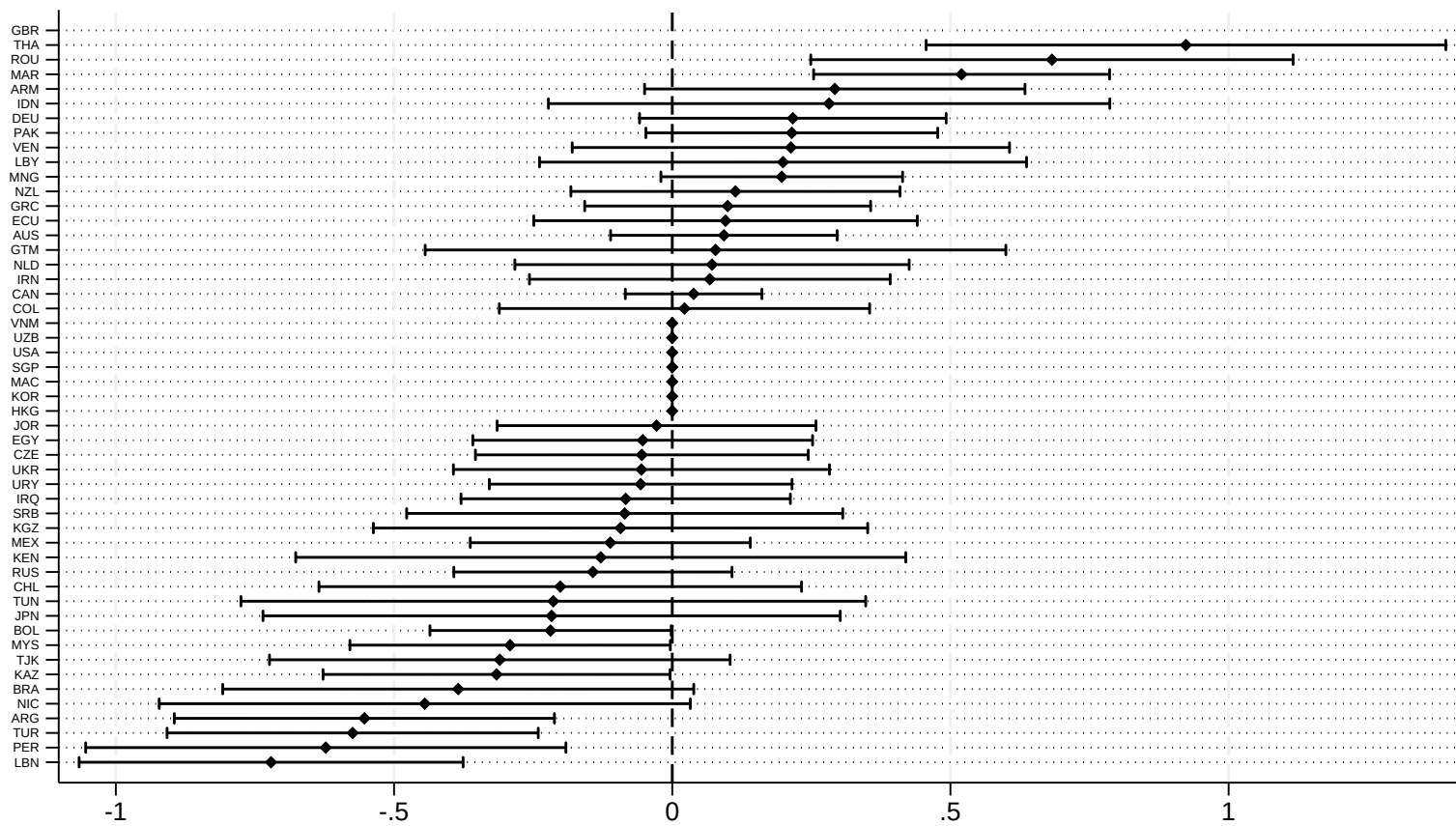


Figure 14



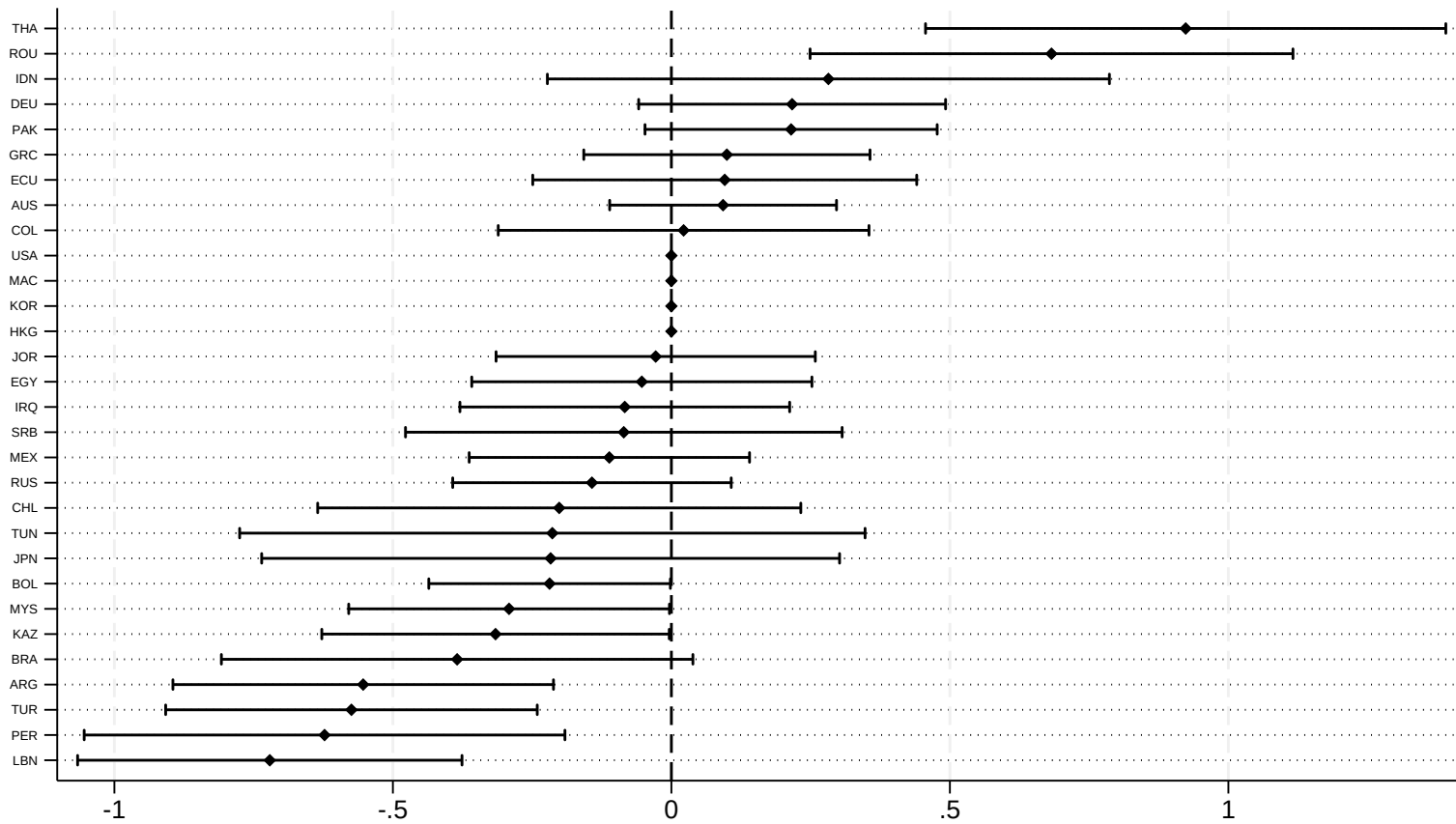
b-coefficients and 95%-CI of life autonomy per country on dummy for lt20k v gt500k (base case)

Figure 15: Dropping age age2 and gender



b-coefficients and 95%-CI of life autonomy per country on dummy for lt20k v gt500k (base case)

Figure 16: Adding education—again, very similar results



b-coefficients and 95%-CI of life autonomy per country on dummy for lt20k v gt500k (base case)

Figure 17: only <2020–fewer countries but still most insignificant.

2.3.2 Interactions with financial satisfaction, class, and income

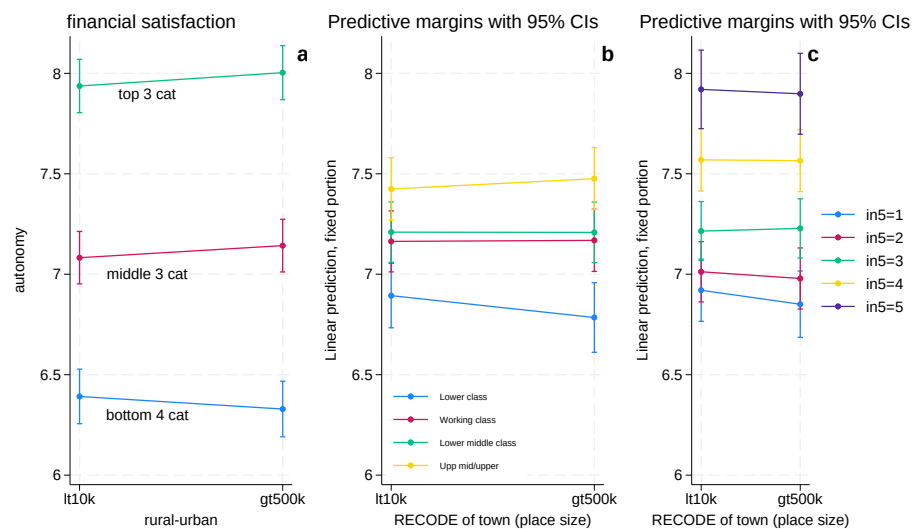


Figure 18: Multilevel country random intercept regressions of freedom on rural-urban dummies interacted with: a) financial satisfaction, b) class, c) income. For underlying regressions see appendix. Controls are the same as in the previous section.

	a	b	c
freedom			
lt10k	0.06	0.11+	0.07
10-100k	-0.07+	0.03	-0.01
100-500k	0.01	-0.05	0.07
gt500k	0.00	0.00	0.00
RECODE of satFin (financial satisfaction)=1	0.00		
RECODE of satFin (financial satisfaction)=2	0.81***		
RECODE of satFin (financial satisfaction)=3	1.67***		
lt10k × RECODE of satFin (financial satisfaction)=1	0.00		
lt10k × RECODE of satFin (financial satisfaction)=2	-0.12*		
lt10k × RECODE of satFin (financial satisfaction)=3	-0.13*		
10-100k × RECODE of satFin (financial satisfaction)=1	0.00		
10-100k × RECODE of satFin (financial satisfaction)=2	0.08+		
10-100k × RECODE of satFin (financial satisfaction)=3	-0.01		
100-500k × RECODE of satFin (financial satisfaction)=1	0.00		
100-500k × RECODE of satFin (financial satisfaction)=2	-0.04		
100-500k × RECODE of satFin (financial satisfaction)=3	-0.06		
gt500k × RECODE of satFin (financial satisfaction)=1	0.00		
gt500k × RECODE of satFin (financial satisfaction)=2	0.00		
gt500k × RECODE of satFin (financial satisfaction)=3	0.00		
age	0.01***	0.01*	0.01*
age2	-0.00***	-0.00	-0.00
male	0.01	0.01	-0.00
married or living together as married	0.01	0.06**	0.04*
Full time	0.00	0.00	0.00
Part time	-0.10***	-0.11***	-0.10***
Self employed	-0.02	-0.04+	-0.03
Retired	-0.06*	-0.03	-0.01
Housewife	-0.21***	-0.20***	-0.18***
Students	-0.14***	-0.12***	-0.11**
Unemployed	-0.14***	-0.22***	-0.18***
Other	-0.27***	-0.32***	-0.30***
health	0.37***	0.50***	0.49***
How many children do you have	0.01*	0.01+	0.01+
religion important	0.04***	0.05***	0.05***
Lower class		0.00	
Working class		0.38***	
Lower middle class		0.42***	
Upp mid/upper		0.69***	
lt10k × Lower class		0.00	
lt10k × Working class		-0.11	
lt10k × Lower middle class		-0.11	
lt10k × Upp mid/upper		-0.16*	
10-100k × Lower class		0.00	
10-100k × Working class		-0.09	
10-100k × Lower middle class		-0.03	
10-100k × Upp mid/upper		-0.04	
100-500k × Lower class		0.00	
100-500k × Working class		0.01	
100-500k × Lower middle class		0.03	
100-500k × Upp mid/upper		0.07	
gt500k × Lower class		0.00	
gt500k × Working class		0.00	
gt500k × Lower middle class		0.00	
gt500k × Upp mid/upper		0.00	
RECODE of inc (income)=1			0.00
RECODE of inc (income)=2			0.13*
RECODE of inc (income)=3			0.38***
RECODE of inc (income)=4			0.71***
RECODE of inc (income)=5			1.05***
lt10k × RECODE of inc (income)=1			0.00
lt10k × RECODE of inc (income)=2			-0.04
lt10k × RECODE of inc (income)=3			-0.08
lt10k × RECODE of inc (income)=4			-0.07
lt10k × RECODE of inc (income)=5			-0.05
10-100k × RECODE of inc (income)=1			0.00
10-100k × RECODE of inc (income)=2			-0.01
10-100k × RECODE of inc (income)=3			0.00
10-100k × RECODE of inc (income)=4			0.02
10-100k × RECODE of inc (income)=5			-0.04
100-500k × RECODE of inc (income)=1			0.00
100-500k × RECODE of inc (income)=2			-0.11
100-500k × RECODE of inc (income)=3			-0.05
100-500k × RECODE of inc (income)=4			-0.14
100-500k × RECODE of inc (income)=5			-0.27*
gt500k × RECODE of inc (income)=1			0.00
gt500k × RECODE of inc (income)=2			0.00
gt500k × RECODE of inc (income)=3			0.00
gt500k × RECODE of inc (income)=4			0.00
gt500k × RECODE of inc (income)=5			0.00
constant	4.51***	4.52***	4.61***
lns1.1.1			
constant	-0.68***	-0.54***	-0.56***
lnsig.e			
constant	0.73***	0.76***	0.76***
N	87874	86391	86471

+ 0.10 * 0.05 ** 0.01 *** 0.001; robust std err

Table 6: OLS regressions of freedom/agency

We could have explored more fully here using usual sequential elaboration of the models, but for now just sticking with the final full model as there are already plenty of robustness/sensitivity checks and one could easily get lost in this sea of extra results.

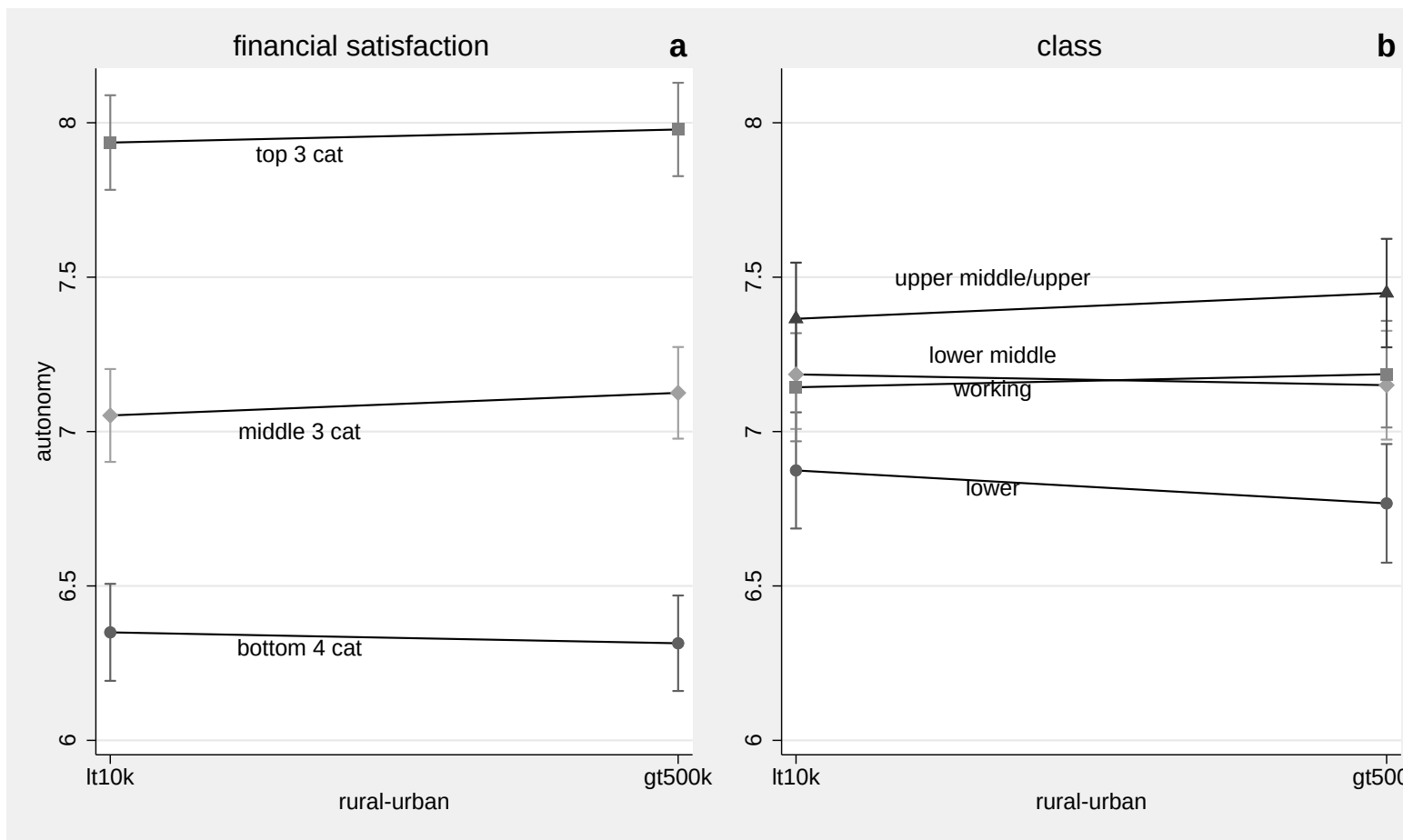


Figure 19: Robustness/sensitivity check: on the sample after drooping countries where top gt500k cat missing. Results are mostly similar to those reported in the body of the paper.

Last we check alternative specification if we are not overcontrolling (Bartram et al. 2024)—in the body of the paper we used a set of control variables from the literature, however, we think that we may drop gender, age and age2—as they may not be predictive of urbanicity. Results are in figure 20 and are substantively the same, and so when adding education in figure 21. The very final check is dropping pandemic years 2020 and later—pandemic might have made city life less free, but our results remain similar, if anything stronger in figure 22

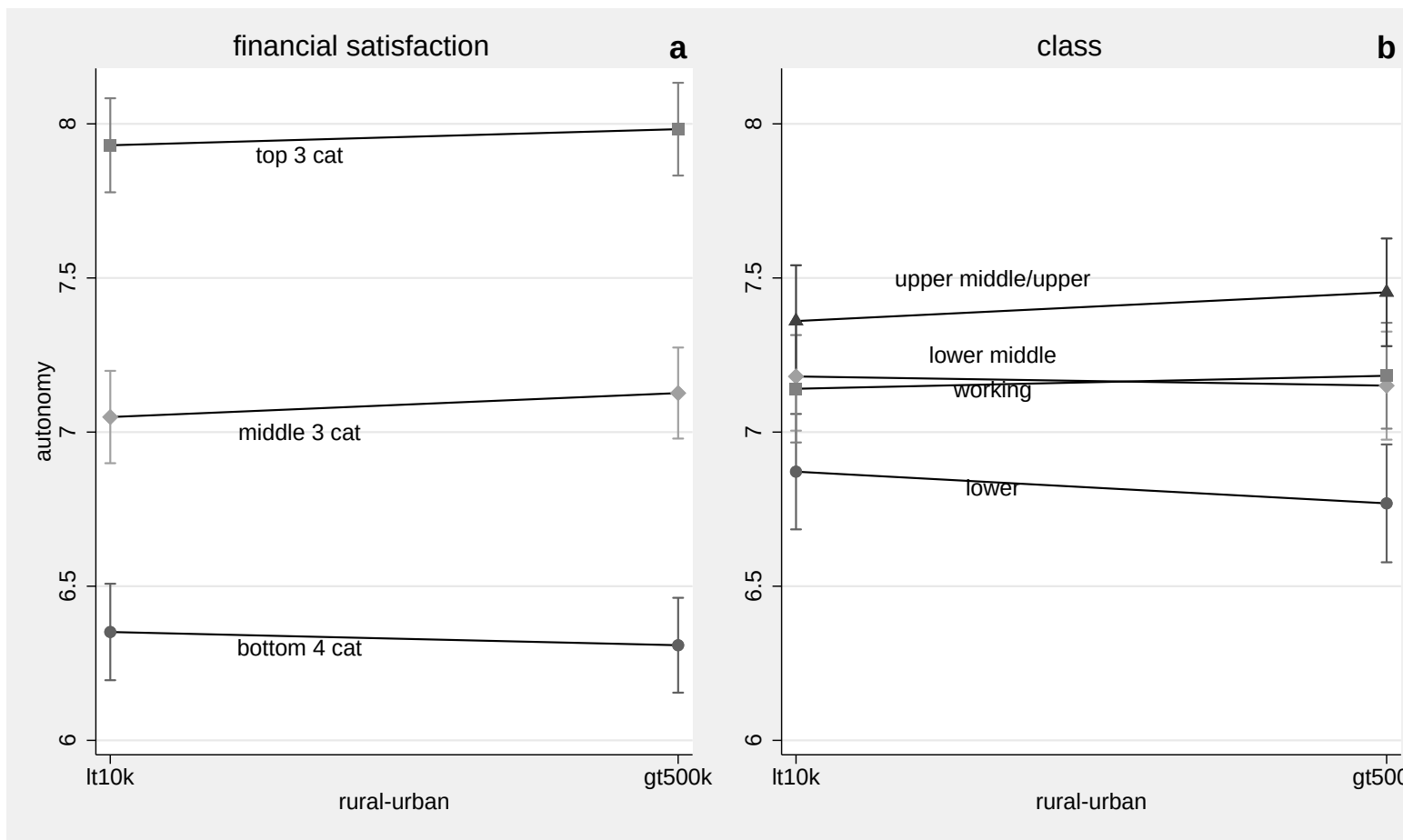


Figure 20: Dropping age age2 and gender.

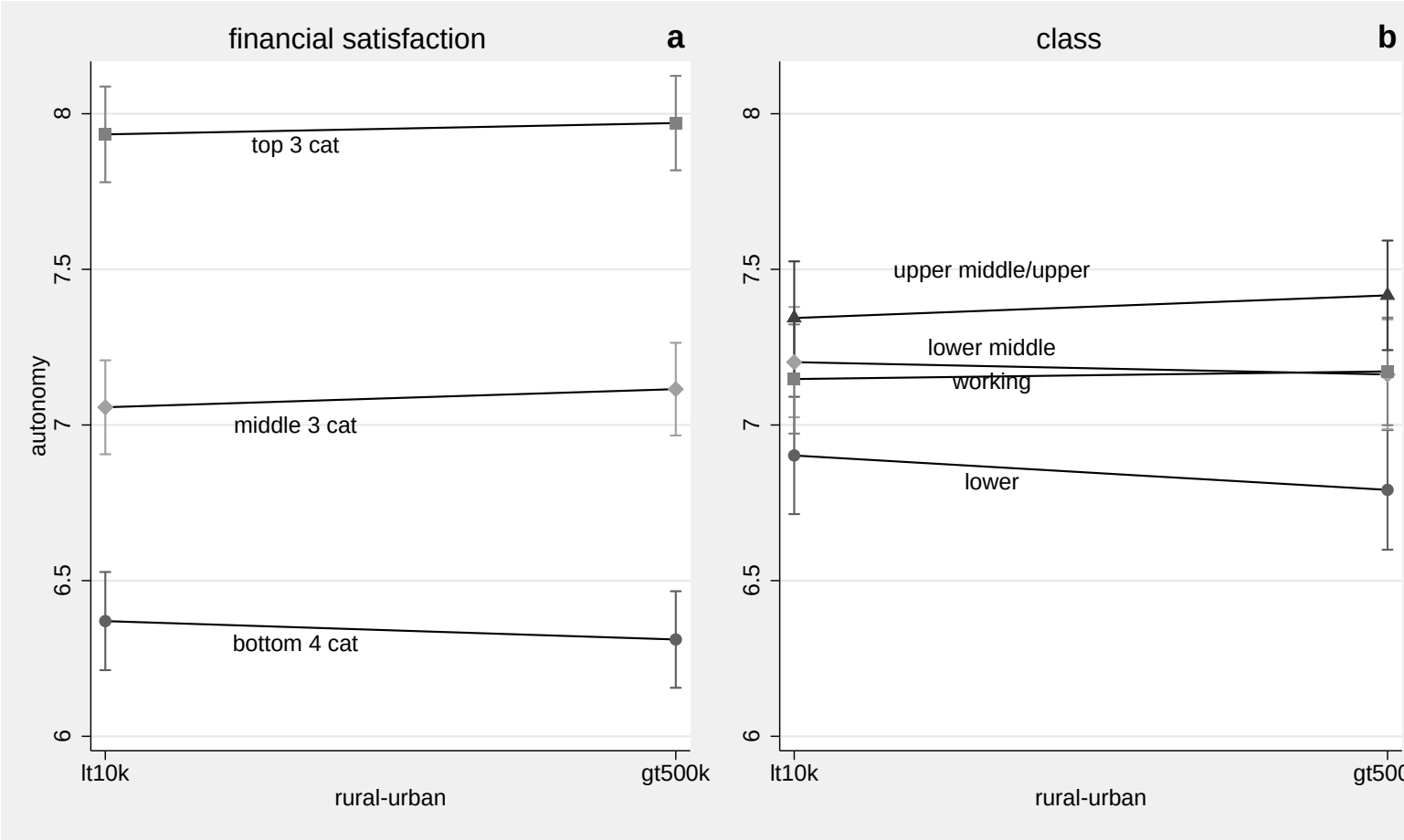


Figure 21: Adding education.

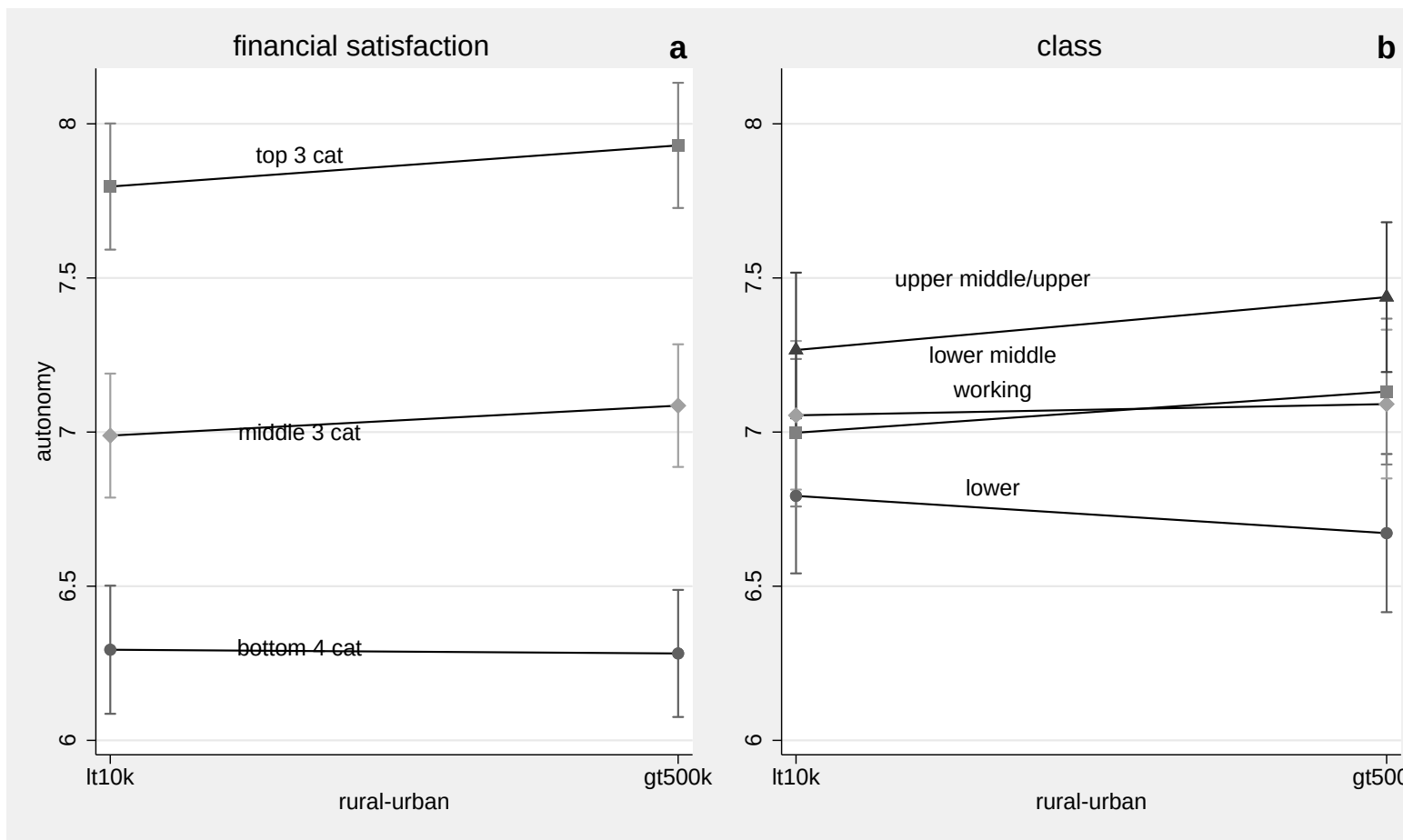


Figure 22: Only <2020.

Initial interactions and alternative specifications

```
. mixed free c.town#c.satFin trust famUns criVic age age2 male mar i.emp he
> alth kids rel_imp || cc: ,mle
Performing EM optimization ...
Performing gradient-based optimization:
Iteration 0: Log likelihood = -184956.1
Iteration 1: Log likelihood = -184956.1
Computing standard errors ...
Mixed-effects ML regression
Group variable: cc
```

```
Number of obs   = 86,280
Number of groups = 64
Obs per group:
    min = 747
    avg = 1,348.1
    max = 4,018
Wald chi2(20)   = 11068.66
Prob > chi2     = 0.0000
```

Log likelihood = -184956.1

free	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
town	-0.03	0.01	-3.99	0.00	-0.05	-0.02
satFin	0.22	0.01	31.95	0.00	0.20	0.23
c.town#						
c.satFin	0.01	0.00	5.90	0.00	0.00	0.01
trust	0.02	0.02	1.34	0.18	-0.01	0.06
famUns	-0.11	0.01	-12.26	0.00	-0.13	-0.09
criVic	-0.11	0.03	-4.22	0.00	-0.16	-0.06
age	0.02	0.00	5.99	0.00	0.01	0.02
age2	-0.00	0.00	-5.37	0.00	-0.00	-0.00
male	0.01	0.02	0.67	0.50	-0.02	0.04
mar	0.00	0.02	0.03	0.98	-0.03	0.03
emp						
Part time	-0.08	0.03	-2.82	0.00	-0.13	-0.02
Self empl..	-0.01	0.02	-0.56	0.57	-0.06	0.03
Retired	-0.04	0.03	-1.32	0.19	-0.10	0.02
Housewife	-0.20	0.03	-8.00	0.00	-0.25	-0.15
Students	-0.14	0.04	-3.79	0.00	-0.21	-0.07
Unemployed	-0.10	0.03	-3.29	0.00	-0.15	-0.04
Other	-0.22	0.07	-3.28	0.00	-0.36	-0.09

health	0.33	0.01	37.13	0.00	0.31	0.35
kids	0.02	0.01	2.67	0.01	0.00	0.03
rel_imp	0.04	0.01	4.11	0.00	0.02	0.05
_cons	4.26	0.11	38.13	0.00	4.04	4.48

Random-effects parameters	Estimate	Std. err.	[95% conf. interval]	
cc: Identity				
var(_cons)	0.28	0.05	0.19	0.39
var(Residual)	4.25	0.02	4.21	4.29

LR test vs. linear model: chibar2(01) = 4378.78 Prob >= chibar2 = 0.0000

.19

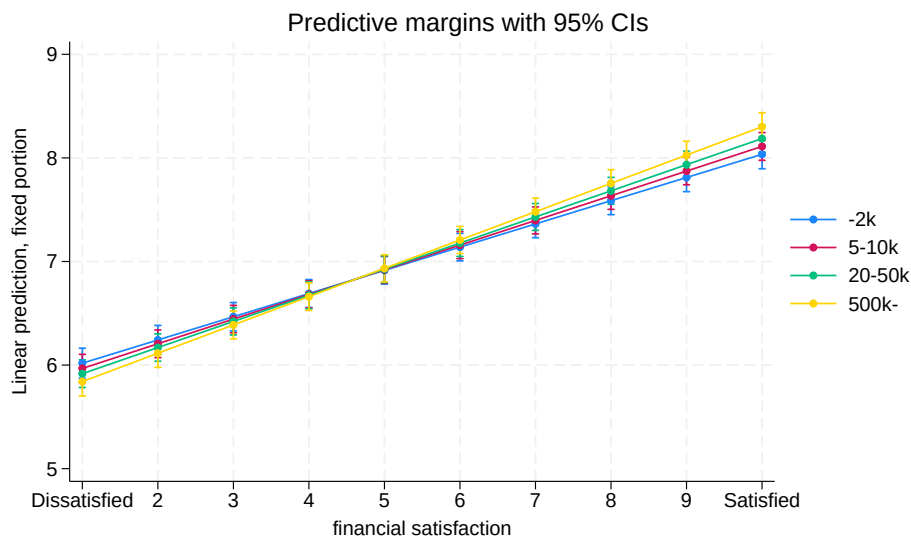


Figure 23: ini1

few things to note in the above—we have trust, famUns and criVic—which are rather predicted by urbanicity, not predictors, hence we may have overcontrol bias. Also here both town and satFin are forced to be linear, a tenuous assumption.

```

mixed free i.t4##i.sf3 trust famUns criVic age age2 male mar i.emp health
> kids rel_imp || cc: ,mle
Performing EM optimization ...
Performing gradient-based optimization:
Iteration 0: Log likelihood = -185241.39
Iteration 1: Log likelihood = -185241.39 (backed up)
Computing standard errors ...
Mixed-effects ML regression
Group variable: cc

```

```

Number of obs   = 86,280
Number of groups = 64
Obs per group:
    min = 747
    avg = 1,348.1
    max = 4,018
Wald chi2(28)   = 10427.08
Prob > chi2     = 0.0000

```

Log likelihood = -185241.39

free	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
t4						
10-100k	-0.14	0.04	-3.39	0.00	-0.21	-0.06
100-500k	-0.04	0.05	-0.78	0.43	-0.13	0.06
gt500k	-0.03	0.04	-0.74	0.46	-0.12	0.05
sf3						
2	0.65	0.03	18.99	0.00	0.59	0.72
3	1.50	0.04	40.61	0.00	1.42	1.57
t4#sf3						
10-100k#2	0.21	0.05	4.38	0.00	0.12	0.31
10-100k#3	0.12	0.05	2.38	0.02	0.02	0.22
100-500k#2	0.09	0.06	1.62	0.11	-0.02	0.20
100-500k#3	0.07	0.06	1.17	0.24	-0.05	0.19
gt500k#2	0.12	0.05	2.43	0.01	0.02	0.22
gt500k#3	0.12	0.05	2.26	0.02	0.02	0.23
trust	0.03	0.02	1.80	0.07	-0.00	0.07
famUns	-0.12	0.01	-13.44	0.00	-0.14	-0.10
criVic	-0.12	0.03	-4.75	0.00	-0.17	-0.07
age	0.02	0.00	5.32	0.00	0.01	0.02
age2	-0.00	0.00	-4.66	0.00	-0.00	-0.00
male	0.01	0.02	0.70	0.48	-0.02	0.04
mar	0.01	0.02	0.34	0.73	-0.03	0.04

emp						
Part time	-0.08	0.03	-3.06	0.00	-0.14	-0.03
Self empl..	-0.02	0.02	-0.70	0.48	-0.06	0.03
Retired	-0.05	0.03	-1.66	0.10	-0.11	0.01
Housewife	-0.22	0.03	-8.41	0.00	-0.27	-0.17
Students	-0.14	0.04	-3.88	0.00	-0.21	-0.07
Unemployed	-0.13	0.03	-4.41	0.00	-0.19	-0.07
Other	-0.25	0.07	-3.65	0.00	-0.39	-0.12
health	0.35	0.01	39.72	0.00	0.34	0.37
kids	0.01	0.01	2.26	0.02	0.00	0.03
rel_imp	0.04	0.01	4.41	0.00	0.02	0.06
_cons	4.82	0.11	45.42	0.00	4.61	5.03

Random-effects parameters	Estimate	Std. err.	[95% conf. interval]
cc: Identity			
var(_cons)	0.27	0.05	0.19 0.39
var(Residual)	4.28	0.02	4.23 4.32

LR test vs. linear model: chibar2(01) = 4250.93 Prob >= chibar2 = 0.0000

.19

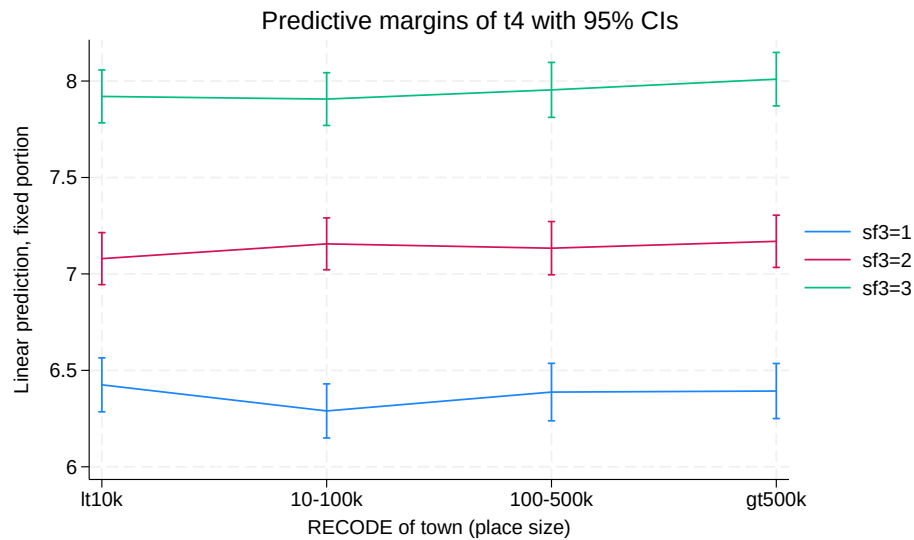


Figure 24: ini2

In the above a different take—financial satisfaction tertiles—for the 1st tertile the biggest drop is actually in the middle category at 10-100k.

```

mixed free c.town#c.inc trust famUns criVic age age2 male mar i.emp healt
> h kids rel_imp || cc: ,mle
Performing EM optimization ...
Performing gradient-based optimization:
Iteration 0: Log likelihood = -184725.92
Iteration 1: Log likelihood = -184725.92 (backed up)
Computing standard errors ...
Mixed-effects ML regression
Group variable: cc
Number of obs = 84,979
Number of groups = 64
Obs per group:
    min = 749
    avg = 1,327.8
    max = 4,018
Wald chi2(20) = 5460.49
Prob > chi2 = 0.0000
Log likelihood = -184725.92

```

free	Coefficient	Std. err.	z	P> z	[95% conf. interval]
town	-0.00	0.01	-0.43	0.67	-0.02 0.01
inc	0.11	0.01	13.01	0.00	0.09 0.12
c.town#c.inc	0.00	0.00	1.21	0.23	-0.00 0.00
trust	0.07	0.02	3.91	0.00	0.04 0.11
famUns	-0.17	0.01	-18.82	0.00	-0.19 -0.16
criVic	-0.16	0.03	-6.05	0.00	-0.21 -0.11
age	0.01	0.00	2.60	0.01	0.00 0.01
age2	-0.00	0.00	-1.15	0.25	-0.00 0.00
male	-0.00	0.02	-0.17	0.87	-0.03 0.03
mar	0.04	0.02	1.97	0.05	0.00 0.07
emp					
Part time	-0.08	0.03	-2.83	0.00	-0.14 -0.02
Self empl..	-0.01	0.02	-0.46	0.65	-0.06 0.04

Retired	0.01	0.03	0.33	0.74	-0.05	0.07
Housewife	-0.18	0.03	-6.65	0.00	-0.23	-0.12
Students	-0.11	0.04	-2.99	0.00	-0.18	-0.04
Unemployed	-0.15	0.03	-5.04	0.00	-0.21	-0.09
Other	-0.25	0.07	-3.54	0.00	-0.39	-0.11
health	0.46	0.01	51.21	0.00	0.45	0.48
kids	0.01	0.01	1.93	0.05	-0.00	0.03
rel_imp	0.05	0.01	5.88	0.00	0.04	0.07
_cons	4.75	0.12	39.88	0.00	4.51	4.98

Random-effects parameters	Estimate	Std. err.	[95% conf. interval]	
cc: Identity				
var(_cons)	0.35	0.06	0.25	0.50
var(Residual)	4.51	0.02	4.47	4.55

LR test vs. linear model: chibar2(01) = 5398.66 Prob >= chibar2 = 0.0000

.19

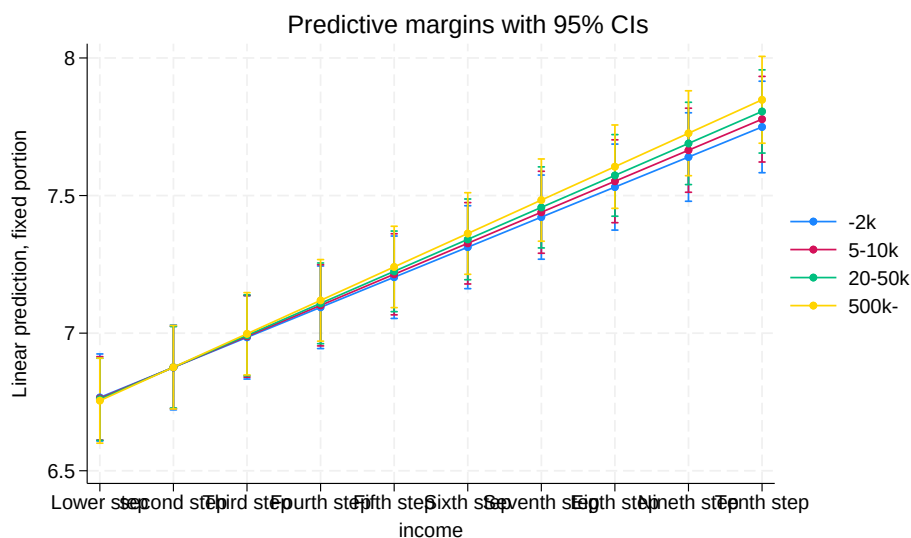


Figure 25: ini3

Here what is easy to see is that along the income spectrum the largest places $> 500k$ change most; and smallest places $< 2k$ change the least

```

mixed free c.town##c.class trust famUns criVic age age2 male mar i.emp hea
> lth kids rel_imp || cc: ,mle
Performing EM optimization ...
Performing gradient-based optimization:
Iteration 0: Log likelihood = -184877.86
Iteration 1: Log likelihood = -184877.86
Computing standard errors ...
Mixed-effects ML regression
Group variable: cc
Number of obs   = 84,931
Number of groups = 64
Obs per group:
    min = 686
    avg = 1,327.0
    max = 4,018
Wald chi2(20)   = 4850.01
Prob > chi2     = 0.0000
Log likelihood = -184877.86

```

	free	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
town		-0.00	0.01	-0.52	0.60	-0.02	0.01
class		0.14	0.02	7.62	0.00	0.10	0.18
c.town#							
c.class		0.01	0.00	1.56	0.12	-0.00	0.01
trust		0.09	0.02	4.79	0.00	0.05	0.13
famUns		-0.18	0.01	-19.22	0.00	-0.20	-0.16
criVic		-0.17	0.03	-6.34	0.00	-0.22	-0.11
age		0.01	0.00	2.45	0.01	0.00	0.01
age2		-0.00	0.00	-1.24	0.22	-0.00	0.00
male		0.01	0.02	0.41	0.68	-0.02	0.04
mar		0.05	0.02	2.96	0.00	0.02	0.09
emp							
Part time		-0.10	0.03	-3.45	0.00	-0.16	-0.04
Self empl..		-0.03	0.02	-1.22	0.22	-0.08	0.02
Retired		-0.02	0.03	-0.64	0.52	-0.08	0.04
Housewife		-0.21	0.03	-7.84	0.00	-0.26	-0.16

Students	-0.13	0.04	-3.46	0.00	-0.20	-0.06
Unemployed	-0.21	0.03	-6.93	0.00	-0.27	-0.15
Other	-0.30	0.07	-4.13	0.00	-0.44	-0.16
health	0.48	0.01	52.29	0.00	0.46	0.49
kids	0.01	0.01	1.81	0.07	-0.00	0.02
rel_imp	0.05	0.01	5.88	0.00	0.04	0.07
_cons	4.86	0.12	39.41	0.00	4.61	5.10

Random-effects parameters	Estimate	Std. err.	[95% conf. interval]
cc: Identity			
var(_cons)	0.36	0.06	0.25 0.51
var(Residual)	4.54	0.02	4.49 4.58

LR test vs. linear model: $\chi^2(01) = 5460.71$ Prob >= $\chi^2 = 0.0000$

.19

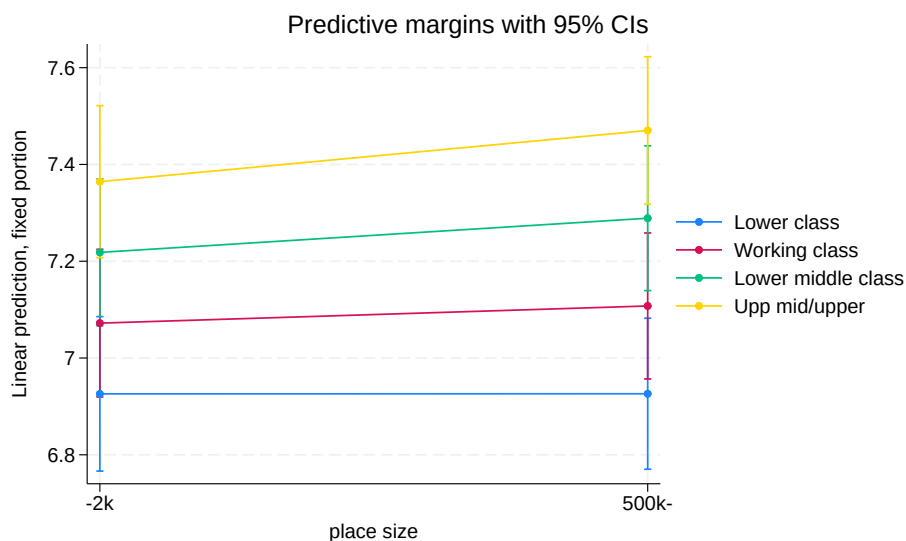


Figure 26: ini4

The above is a version of the main graph from the body of the paper, again here we are over-controlling with trust and the 2 crime variables, but it is not that different after all, similar patterns, just the main difference being that here the lower class is flat and in the body downward sloping and here the top class is quite steep and while in paper it is upward sloping it's little less so in the body of the paper

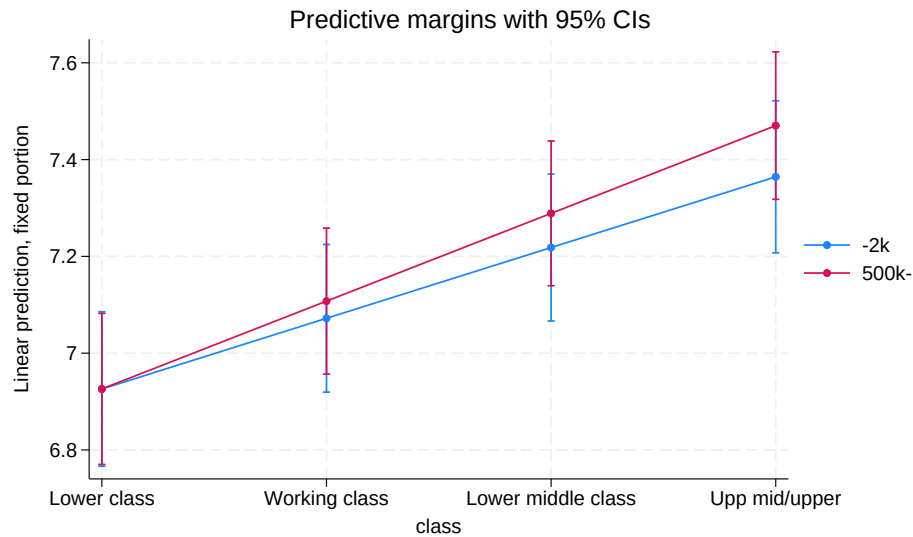


Figure 27: ini5

ini5 is from the same model, nicely shows that the higher the class the more difference between the smallest and the largest places

```

mixed free c.town##i.class trust famUns criVic age age2 male mar i.emp hea
> lth kids rel_imp || cc: ,mle
Performing EM optimization ...
Performing gradient-based optimization:
Iteration 0: Log likelihood = -184853.93
Iteration 1: Log likelihood = -184853.93 (backed up)
Computing standard errors ...
Mixed-effects ML regression
Group variable: cc
Number of obs = 84,931
Number of groups = 64
Obs per group:
    min = 686
    avg = 1,327.0
    max = 4,018
Wald chi2(24) = 4900.64
Prob > chi2 = 0.0000
Log likelihood = -184853.93

```

	free	Coefficient	Std. err.	z	P> z	[95% conf. interval]
town		-0.00	0.01	-0.54	0.59	-0.02 0.01
class						
Working c..		0.22	0.06	3.86	0.00	0.11 0.34
Lower mid..		0.28	0.06	5.10	0.00	0.17 0.39
Upp mid/u..		0.48	0.06	7.75	0.00	0.36 0.60
class#c.town						
Working c..		0.01	0.01	1.34	0.18	-0.01 0.03
Lower mid..		0.01	0.01	1.28	0.20	-0.01 0.03
Upp mid/u..		0.02	0.01	1.95	0.05	-0.00 0.04
trust		0.09	0.02	4.82	0.00	0.05 0.13
famUns		-0.18	0.01	-19.18	0.00	-0.20 -0.16
criVic		-0.16	0.03	-6.30	0.00	-0.22 -0.11
age		0.01	0.00	2.50	0.01	0.00 0.01
age2		-0.00	0.00	-1.29	0.20	-0.00 0.00
male		0.01	0.02	0.35	0.72	-0.03 0.04
mar		0.05	0.02	2.87	0.00	0.02 0.09
emp						
Part time		-0.10	0.03	-3.39	0.00	-0.15 -0.04
Self empl..		-0.03	0.02	-1.10	0.27	-0.07 0.02
Retired		-0.02	0.03	-0.52	0.60	-0.08 0.05
Housewife		-0.20	0.03	-7.60	0.00	-0.25 -0.15
Students		-0.13	0.04	-3.34	0.00	-0.20 -0.05
Unemployed		-0.20	0.03	-6.63	0.00	-0.26 -0.14
Other		-0.29	0.07	-4.01	0.00	-0.43 -0.15
health		0.47	0.01	52.12	0.00	0.46 0.49
kids		0.01	0.01	1.85	0.06	-0.00 0.03
rel_imp		0.05	0.01	5.85	0.00	0.04 0.07
_cons		4.96	0.12	40.78	0.00	4.72 5.20

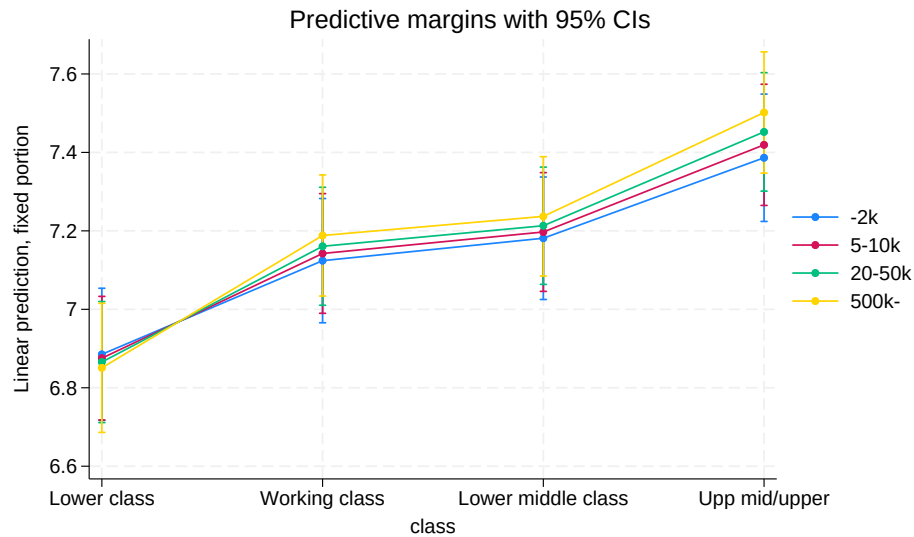


Figure 28: ini6

And then ini6: couple interesting things: first things flip from lower to working class, then working and lower middle class quite similar, and the differences get larger for upper middle and upper classes.

2.3.3 By region and country income group: Interactions with financial satisfaction

Here we repeat urbanicity interactions with financial satisfaction from the body of the paper but by region and income group; the first panel in each graph is repeated from the body of the paper.

In figure 29 we see that the pattern is much stronger for East Asia/Pacific, and opposite especially in Middle East/North Africa, but also in North America.

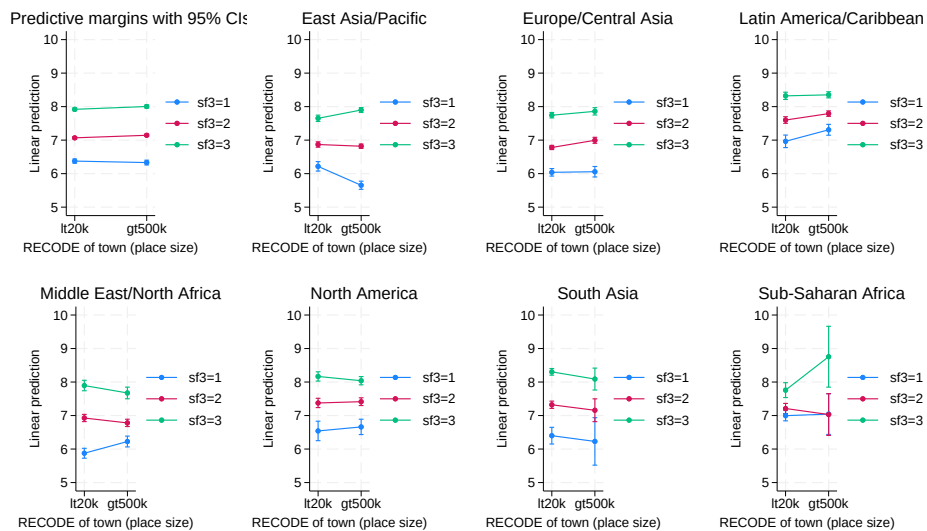


Figure 29: Urbanicity interactions with financial satisfaction by region.

In figure 30 high income has a similar pattern, and lower middle income panel has a strong pattern, and quite the opposite pattern for upper middle income.

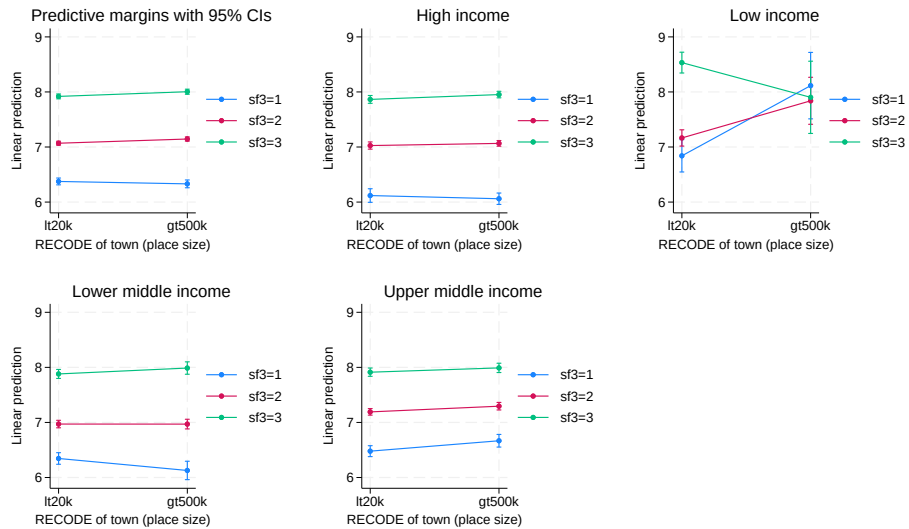


Figure 30: Urbanicity interactions with financial satisfaction by income group. Note just 2 countries for low income group: ETH TJK.

3 Auxiliary results and sidenote

3.1 Some graphs

fig 31 and 32

weird to think that more ineq and pov would produce more freedom/autonomy; rather i guess lax capitalist friendly regulation would produce more ineq and pov, but should also produce less autonomy/freedom on avg, but it doesn't, there is positive relationship

or maybe the other way round: more freedom/autonomy more ineq and pov

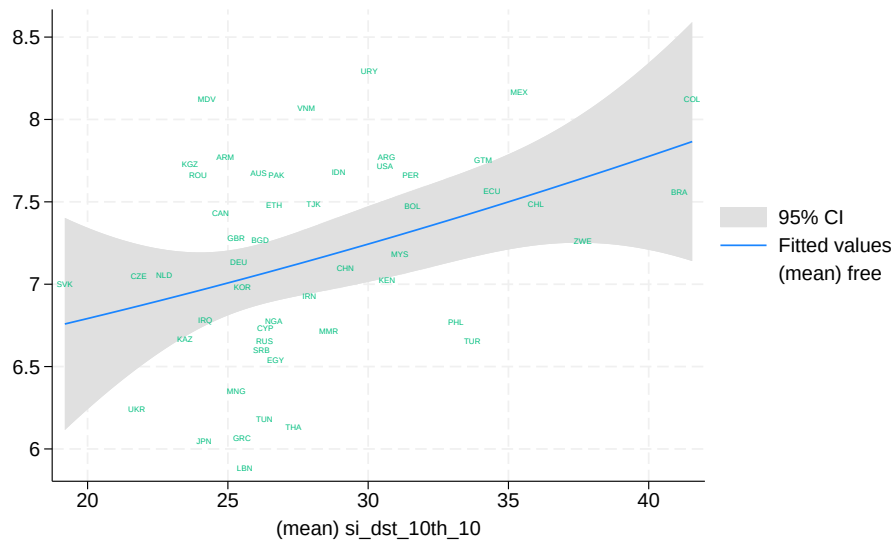


Figure 31: more ineq (inc share held by top 10perc, more autonomy-conservatives would like it!! and pretty strong corr .4

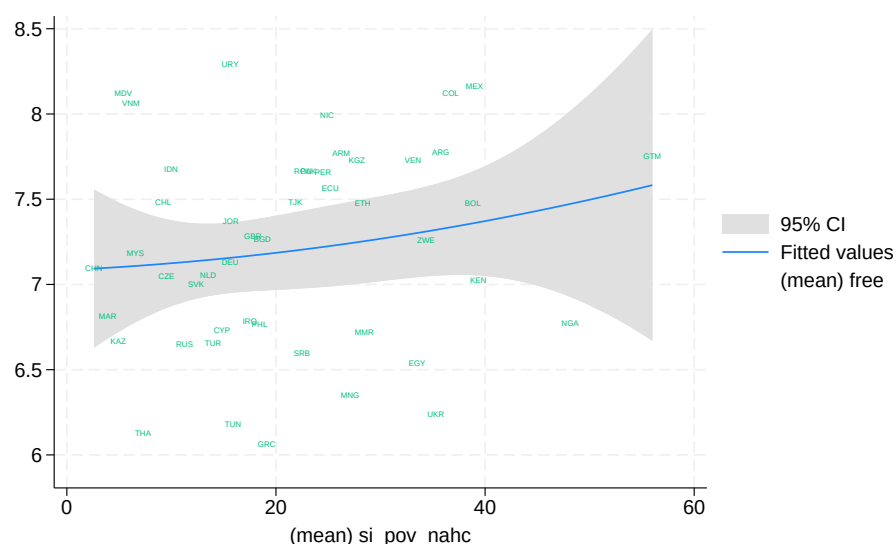


Figure 32: likewise with poverty, but here corr just .2

3.2 Literature

Newman (2020) makes some useful points that may be worth citing in full:

Social theory typically conceives of urban growth as associated with political freedom. [...] But not even in the most likely of cases did the city makes urban peasants free. Focusing on Mexico City, I argue that because rural-to-urban migrants became urban denizens by establishing squatter settlements, they entered into dependent relations with political elites, which reinforced a system of political dependency, or clientelist politics. [...] Mexico is, perhaps, the most iconic case. A large cross section of the peasantry rose in arms against landowning oligarchs in the insurrectionary phase of the Mexican Revolution (1910-1917), ultimately precipitating the wholesale reconstruction of the political order. Thereafter, peasants continued to mobilize for radical change, constituting a major obstacle to postrevolutionary governing elites who sought to consolidate control. [...] When a large and growing portion of rural-to-urban migrants live in squatter settlements, and therefore have little recourse to legal protection in residential matters, they almost inevitably grow dependent on political elites—in an arrangement known as “clientelism.” Urban concentration gives rise to clientelism.

```
. tabstat ls, by(lwd7) stat(mean sd) format(%9.1f)
```

Summary for variables: ls
Group variable: lwd7 (RECODE of lwd (do what you like))

lwd7	Mean	SD
0-4	5.6	2.7
2	7.6	2.3
3	7.6	1.8
4	8.0	1.5
5	7.8	1.9
6	8.2	1.5
7	8.9	1.5
Total	8.3	1.9

.9

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